

Consumption Response to Anticipated Income Changes: Evidence from the Magnitude Effect*

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Abstract

We study how household consumption responds to a positive and anticipated income shock: the completion of fixed-schedule car loan repayments. Using novel individual-level panel data from the Bank of Korea, we document a magnitude effect in which the behavioral response depends critically on the size of the anticipated income change. In response to small payments, consumption exhibits pronounced excess sensitivity, with marginal propensities to consume (MPCs) of 0.76. In contrast, large payments yield MPCs as low as 0.14, consistent with near-optimal smoothing. This nonlinear pattern persists across age, income, and liquidity groups, and cannot be explained by liquidity constraints alone. We interpret these findings through the lens of bounded rationality, where agents selectively optimize consumption only when the welfare stakes and utility gains are sufficiently high. Our results offer valuable insights into predicting household consumption responses to government policy interventions.

JEL Classification: D12, E21, G11

Keywords: Anticipated income changes, Consumption, Heterogeneity

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1 Introduction

How households adjust consumption in response to income fluctuations is a fundamental question in macroeconomics and household finance, with direct implications for the effectiveness of policies that operate through predictable income components. Under the life-cycle permanent income hypothesis (LCPIH), a fully rational, forward-looking household is expected to smooth consumption over time; consequently, predictable changes in income should have no discernible impact on spending at the moment of realization (Alessie and Lusardi, 1997; Jappelli and Pistaferri, 2010; Fuchs-Schündeln and Hassan, 2016). Despite this theoretical benchmark, a substantial empirical literature documents “excess sensitivity”, a phenomenon wherein households significantly alter their spending in response to foreseeable income shifts.

Two broad mechanisms have been proposed to explain this departure from the standard model. The first emphasizes structural frictions, such as liquidity constraints and limited credit access, which may prevent households from smoothing consumption even when they possess perfect foresight (Carroll, 1997; Parker et al., 2013; Kaplan and Violante, 2014; Baker et al., 2020; Attanasio, Kovacs and Moran, 2024). The second points to behavioral mechanisms, including inattention and mental accounting, suggesting that households may cognitively underreact to anticipated changes until cash is actually received (Baugh et al., 2021; Graham and McDowall, 2026).

While both channels have received substantial attention, considerably less is known about how the *magnitude* of an anticipated income change shapes consumption responses. Scholnick (2013) provides early evidence on this possibility using a small sample of mortgage completions, but the limited sample size and restricted range of income changes prevent general conclusions. Beraja and Zorzi (2024) document related size-dependent patterns in durable spending but focus on a structural model rather than reduced-form heterogeneity across individual characteristics. The question of whether and how the degree to which anticipated income changes are incorporated into consumption decisions varies systematically with their size therefore remains open.

This paper fills the gap in this literature by providing novel evidence on size-dependent excess sensitivity. We exploit a natural experiment in South Korea in which both the timing and magnitude of income gains are fully predetermined: the scheduled completion of fixed-term car loans.¹ Korean

¹This setting closely parallels the identification strategy of Stephens Jr (2008), who exploits U.S. vehicle loan completions to examine excess sensitivity, and complements Andersen, Bech and De Stefani (2024), who utilize predictable mortgage repayment changes using Danish administrative data.

automobile loans are fully amortizing contracts with repayment schedules set at origination (36 to 60 months), and borrowers receive advance monthly notifications of their upcoming completion date, eliminating any informational uncertainty about timing or amount. Crucially, the size of the resulting income gain is determined entirely by initial contract parameters (vehicle price, down payment, and loan duration) rather than contemporaneous financial conditions, rendering the final payment amount effectively orthogonal to financial circumstances at completion. We confirm this property empirically: payment size and quarterly income exhibit a low correlation (0.20), and the income distribution is broadly uniform across payment size terciles. This quasi-exogenous variation in windfall magnitude allows us to isolate the effect of anticipated income change size on spending behavior within a constant institutional environment.

Our analysis employs individual-level data from the Bank of Korea (BOK) Household Debt Database, a stratified panel covering 2.4 percent of the credit-active population. The database contains detailed repayment schedules including payment size and duration, quarterly income, and credit card expenditure derived from financial transactions across all domestic institutions. Within this panel, our estimation sample captures 77,148 loan-completion events from 2012 to 2016, with final payments ranging from \$89 to \$5,660 in US dollars per quarter. This range offers the wide, continuous variation in anticipated income changes necessary to identify size-dependent excess sensitivity. Although institutional loan approval requirements lead to a modest underrepresentation of car loan holders in the lowest income bracket, the estimation sample remains highly representative of the broader credit-active population.

To test excess sensitivity, we compare a treatment group of individuals completing their final loan payment in a given quarter against a control group whose subsequent loan maturities require continued scheduled payments throughout the observation window. Our baseline specification regresses quarterly consumption expenditure on distributed leads and lags of the final payment amount to evaluate both the presence of excess sensitivity and its dynamic profile. The regression model also includes individual, time, and region fixed effects to absorb time-invariant heterogeneity and aggregate shocks. Recognizing that our data reflect spending rather than overall consumption, we refer to our estimated coefficients as measures of the marginal propensities to consume (MPCs) out of the anticipated income change. To examine size dependence, we interact payment amount with distributional groups defined by three measures: the absolute dollar value of the final payment, payment size relative to quarterly before-tax income, and payment size relative to typical quarterly

expenditure. This framework allows us to assess how both absolute and relative income changes shape consumption adjustments, and to determine which measure most powerfully explains the observed heterogeneity in excess sensitivity.

Our main findings are threefold. First, the average MPC out of anticipated income increases is approximately 0.18 and statistically significant, consistent with the range of 0.12 to 0.43 documented in prior studies.² Second, the spending response concentrates sharply at the moment of income realization, with no discernible anticipatory adjustment in the preceding period. This pattern is inconsistent with forward-looking consumption smoothing and is instead indicative of a discrete behavioral response triggered by the receipt of cash. Third, MPC declines monotonically as the size of the income change increases. When measured in absolute terms, individuals in the bottom tercile of the payment distribution exhibit an MPC of 0.76, falling sharply to 0.14 in the top tercile. This heterogeneity is most pronounced when payments are expressed as a share of quarterly income, where estimates range from 0.85 to 0.11, while scaling relative to typical consumption expenditure yields the least explanatory power.

Motivated by the observed heterogeneity in average MPCs, we further decompose consumption responses across the joint distribution of payment size, age, income, and liquidity to isolate the drivers of excess sensitivity.³ Our analysis reveals two consistent findings: first, MPCs decline as the anticipated income change rises relative to baseline income or expenditure, a pattern that holds across all demographic and financial subgroups. Second, while lower-income households tend to have higher MPCs within a given payment size category, variation in payment size generates considerably more dispersion in MPCs than variation in income does. Specifically, moving from the lowest to the highest payment size tercile produces a larger decline in the MPC than an equivalent shift across income groups, holding payment size constant. This asymmetry indicates that payment magnitude is an independent and quantitatively dominant predictor of consumption adjustment, one that cannot be attributed to the liquidity channel alone.

We interpret these findings through the lens of bounded rationality and the welfare economics of consumption smoothing. When the utility loss from suboptimal adjustment is small, households

²See [Shapiro and Slemrod \(2009\)](#), [Parker et al. \(2013\)](#), [Misra and Surico \(2014\)](#), [Kueng \(2018\)](#), and [Coibion, Gorodnichenko and Weber \(2020\)](#) for estimates of the MPC out of predictable income changes in this range. Table A.1 provides a detailed review.

³In the absence of direct wealth data, we follow [Coibion, Gorodnichenko and Weber \(2020\)](#) by proxying liquidity through income levels and mortgage status, the latter serving as a tractable indicator of borrowing constraints.

have little incentive to devote cognitive resources to planning a response to an anticipated income change. Our results are consistent with this mechanism: for small income changes, the welfare cost of inaction is modest and households behave as though the windfall goes unnoticed, generating high MPCs at realization. For large ones, the cost of failing to smooth is sufficient to motivate deliberate forward-looking adjustment, yielding lower MPCs. This mechanism provides a coherent and parsimonious account of the monotonically declining MPC profile we document, and offers a behavioral complement to purely structural explanations of excess sensitivity based on liquidity constraints.

A counterfactual policy experiment reinforces the practical relevance of these findings. Existing interventions such as tax rebates or stimulus checks typically allocate payments based on income thresholds, providing uniform transfers within brackets and obscuring the independent role of payment size. Our setting offers a key advantage: natural variation in the magnitude of anticipated income changes across individuals with differing characteristics, allowing us to isolate and quantify the size effect. We simulate two fiscal transfer scenarios: one that concentrates larger payments on the lowest income tercile, and one that distributes smaller average payments more broadly across the population. Accounting for the size-dependent MPC, aggregate consumption growth rises from 0.47 percent under the concentrated policy to 1.38 percent under the dispersed one despite equivalent aggregate outlays, demonstrating that the magnitude profile of transfers, not merely their total value, shapes macroeconomic effectiveness.

This paper relates to several strands of the literature. First, we add to the growing body of work using large-scale administrative data to study consumption responses to income shocks, including dividend payouts, tax rebates, and stimulus payments ([Hsieh, 2003](#); [Kueng, 2018](#); [Coibion, Gorodnichenko and Weber, 2020](#)), as well as more recent work documenting meaningful spending responses to fully predictable cash flow changes ([Scholnick, 2013](#); [Andersen, Bech and De Stefani, 2024](#)). Second, we contribute to the literature on framing, timing, and payment structure as determinants of household spending ([Shin, Kim and Koh, 2021](#); [Bonaccorsi et al., 2024](#); [Pauls and Laudi, 2025](#)). We extend this strand by showing that payment magnitude operates as an independent dimension of transfer design, shaping MPCs in ways not attributable to income, liquidity, or framing alone. Third, our findings speak to the literature on bounded rationality and behavioral departures from the LCPIH ([Browning and Collado, 2001](#); [Hsieh, 2003](#)). We document robust size-dependent heterogeneity in MPCs irreducible to liquidity constraints, providing empir-

ical grounding for bounded rationality as a micro-foundation of consumption behavior in which welfare loss varies systematically with income-change size. Together, these contributions suggest that the magnitude of an income change is an essential consideration for both economic theory and policy design.

The remainder of the paper is organized as follows. Sections 2 and 3 describe the data and econometric methodology. Section 4 presents the estimation results, Section 5 discusses the underlying mechanisms, and Section 6 evaluates the policy implications. Section 7 concludes.

2 Data and Institutional Background

2.1 Data

Our primary data source is the BOK household debt database, a longitudinal quarterly panel of de-identified individual-level records drawn from a major credit bureau in South Korea. The database is constructed using stratified random sampling calibrated to match the national distribution of age, region, and credit rating, covering approximately 2.4 percent of the credit-active population and yielding a full panel of 896,000 observations. Due to the availability of the database and limitations on data access, our analysis covers the period from 2012 to 2016.⁴

The panel contains four categories of information central to our analysis. First, it records the complete repayment schedule for each individual debt instrument, such as the type of debt, installment size, and payment duration, which allows us to precisely identify both the timing and magnitude of the anticipated income change at loan completion. Second, consumption expenditure is derived from actual credit and debit card transactions aggregated across all major financial institutions, capturing both durable and non-durable spending. Over the sample period, card-based transactions account for 75 percent of total household consumption on average (Credit Finance Association of Korea), and their high co-movement with aggregate consumption growth makes them a reliable proxy for total household spending. Third, before-tax income is collected by the credit bureau for tax-reporting purposes based on verified income documentation, making it less susceptible to the measurement error that affects self-reported survey income. Fourth, a rich set of

⁴The BOK household debt database became available from 2012 onwards; our access covers 2012 to 2016. A longer panel would permit analysis of additional anticipation periods and a wider event window, which we view as a promising direction for future research.

credit variables is available, including credit grade, utilization rate, card liability, and default risk, alongside standard demographic characteristics.

Two limitations of the dataset are worth noting. It does not record asset or wealth holdings, which we partially address by proxying for liquidity constraints using quarterly income, credit utilization rates, and mortgage debt status, following [Fuchs-Schündeln and Hassan \(2016\)](#) and [Coibion, Gorodnichenko and Weber \(2020\)](#); we discuss the implications in Section 4.3. Cash transactions are also unobserved, though given the exceptionally high rate of card usage in South Korea, any resulting attenuation bias would work against finding the excess sensitivity we document.

2.2 The South Korean Auto Loan Market

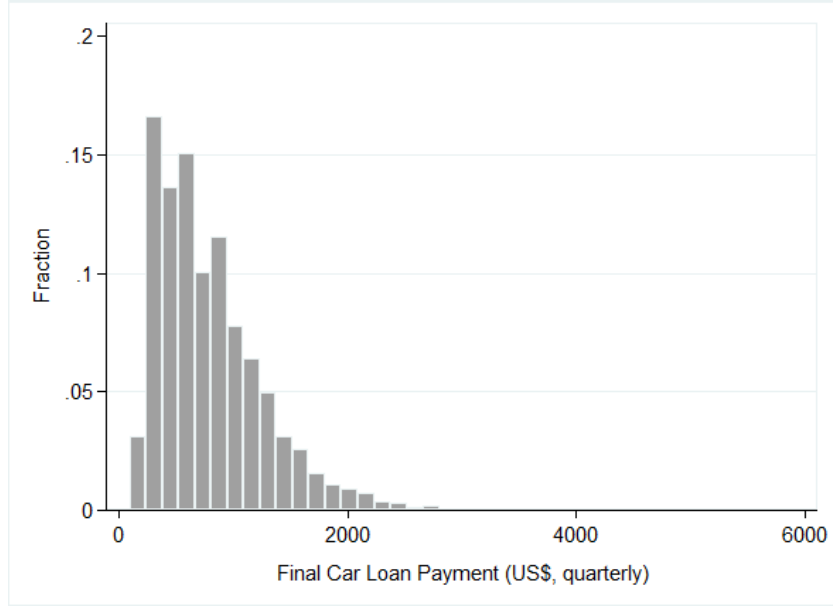
South Korea has a well-developed auto loan market in which installment financing accounts for approximately 70 percent of total vehicle purchases. Domestic vehicle sales grew steadily from approximately 1.41 million units in 2012 to a peak of around 1.82 million units in 2016 over our sample period. The market is served by three broad categories of lenders: captive finance companies affiliated with automakers, commercial banks, and non-bank financial institutions, with commercial banks supplying around 60 percent of automobile loans.

Korean auto loans are fully amortizing, fixed-term contracts with tenors typically ranging from 36 to 60 months. They are originated at the point of vehicle purchase and repaid in equal monthly installments. Loan amounts range from 10 to 90 percent of the vehicle’s value, depending on vehicle price, down payment size, and contract duration, generating substantial heterogeneity in monthly payment obligations across borrowers. Although auto loans account for only around 9 percent of total household debt compared to 54 percent for mortgages, this heterogeneity in payment size is precisely what makes them well-suited to our empirical strategy. Mortgage and student loan completions, by contrast, tend to cluster at similar magnitudes within demographic groups, limiting their usefulness for identifying consumption responses to income changes of varying size.

2.3 Sample Selection and Descriptive Statistics

We construct the estimation sample by imposing three restrictions on the full panel. We retain only individuals holding a single auto loan with regular fixed-term repayments scheduled through to maturity. Early repayers are excluded, as lump-sum prepayment reflects financial decisions endogenously tied to current consumption behavior. Individuals with multiple concurrent auto

Figure 1: Distribution of Final Car Loan Payment, 2012–2016



Notes: This figure displays the distribution of quarterly final car loan payments in US dollars (CPI adjusted) with the base year of 2020. Each bin is \$300 wide.

loans are also excluded to avoid confounding from overlapping loan cycles. Finally, we trim the top and bottom one percent of the final payment distribution to limit the influence of outliers. These restrictions yield a final estimation sample of 77,148 loan completion events.

The resulting payment size distribution, shown in Figure 1, with a mean final payment of \$788 and values ranging from \$89 to \$5,660, each CPI-adjusted to 2020 prices and converted from Korean won to US dollars using average exchange rates. More than half of all observations fall below \$1,000, though the upper tail extends considerably, reflecting heterogeneity in vehicle price and loan terms across borrowers.

Table 1 reports descriptive statistics for the main variables. The mean anticipated income gain at loan completion represents approximately 10 percent of quarterly before-tax income and 25 percent of two-quarter lagged consumption expenditure. Mean quarterly income is \$8,841 and mean quarterly credit card expenditure is \$4,802, each with considerable cross-sectional dispersion. The card utilization rate averages 27 percent, and the sample exhibits relatively strong credit standing with a mean credit grade of 3.30 on a scale of 1 (best) to 10 (worst). The age profile is concentrated among mid-career individuals, with 56 percent of borrowers aged between 40 and 59,

Table 1: Descriptive Statistics

	Mean	Median	St.Dev.
Car Loans			
Quarterly payments	788	682	475
per quarterly before-tax income	9.91%	8.21%	6.61%
per quarterly total expenditures	25.27%	17.66%	24.40%
Quarterly expenditures			
Credit card expenditure (CCE)	4,802	4,091	3,247
Card utilization rate	27.39%	16.84%	58.80%
Quarterly before-tax Income	8,841	8,487	3,231
Card Holders' Characteristics			
Credit grade (scale 1 to 10)	3.30	3.00	2.06
Age between 40 and 59 (%)		56.51%	
Number of observations		77,148	

Notes: All monetary values are in real U.S. dollars, deflated to 2020 prices. Credit grade ranges from 1 to 10, where 1 denotes the highest creditworthiness and 10 the lowest.

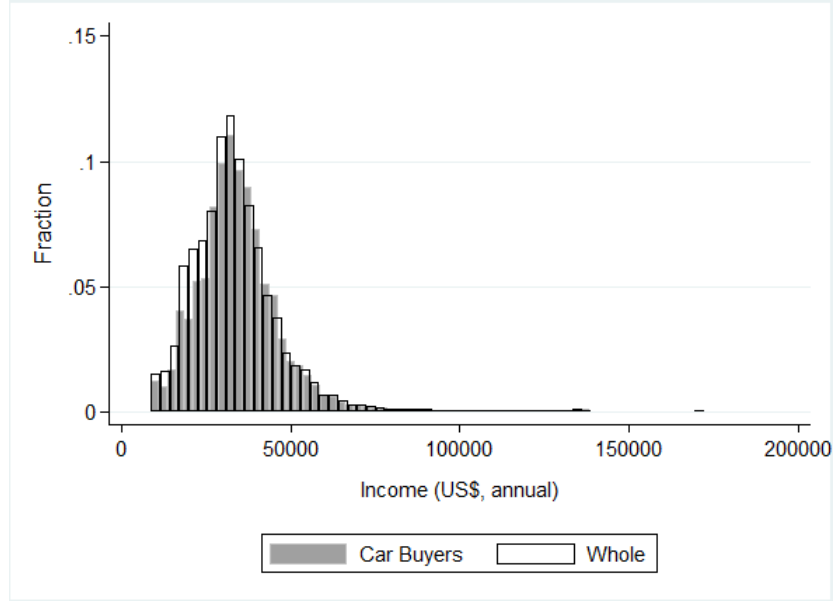
consistent with this cohort's dominance in the Korean auto loan market.

A natural concern with a sample conditioned on auto loan take-up is whether it adequately represents the broader credit-active population. Figure 2 addresses this directly. Panel (a) shows that the annual income distribution of car buyers closely tracks the full credit bureau panel, with a mean annual income of \$35,360 broadly comparable to the full sample; the car buyer group is modestly underrepresented below \$30,000 in annual income, attributable to standard creditworthiness requirements for loan approval. Panel (b) shows a similar alignment in quarterly consumption expenditure: while car buyers exhibit a slightly lower concentration at the bottom of the spending distribution, the overall shape and range remain broadly comparable to the full sample. Additional distributional comparisons, including payment size relative to income and consumption and the age distribution, are reported in Appendix B.

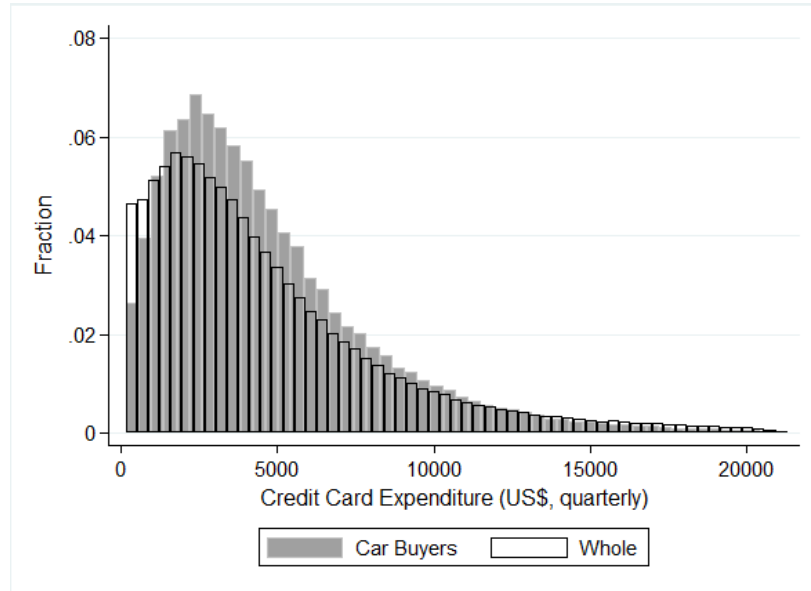
We nonetheless acknowledge that auto loan take-up is unlikely to be fully random. The creditworthiness threshold for loan approval excludes the most financially vulnerable households, and unobserved characteristics such as liquid asset holdings may differ systematically between car buyers and the broader population. To the extent that selection removes those most likely to exhibit high MPCs for structural reasons, however, it works against rather than for a liquidity-constraint interpretation of our findings.

Figure 2: Representativeness: Income and Consumption

(a) Distribution of Annual Income



(b) Distribution of Credit Card Expenditure



Notes: Panel (a) plots the distribution of annual income and panel (b) the distribution of quarterly credit card expenditure, both in real U.S. dollars deflated to 2020 prices. Each bin width is \$1,000 for income and \$300 for expenditure. Shaded bars correspond to car buyers; outlined bars correspond to the full sample from 2012 to 2016.

3 Empirical Methodology

3.1 Baseline Specification

To assess whether consumption responds to the predictable income gain at loan completion, we estimate the standard parametric regression, which is given by:

$$\Delta c_{it} = \alpha_t + \gamma_i + Region_i + \sum_{s=n}^m \beta_s \cdot FP_{i,t-s} + \lambda' x_{it} + \epsilon_{it}, \quad (1)$$

where c_{it} measures the quarterly credit card expenditure for individual i in period t , $FP_{i,t}$ denotes the dollar value of the final car loan payment made by individual i at time t , and $s \in [n, m]$ indexes periods relative to loan completion allowing for leads and lags. The coefficient, β , measures the excess sensitivity to the anticipated income gain at loan completion ($s \geq 0$) and any pre-completion adjustment in spending in anticipation of the payment relief ($s < 0$).⁵ Time, individual, and region fixed effects are captured by α_t , γ_i , and $Region_i$, respectively. Control variables, x_{it} , include demographic characteristics (age dummies, gender), income changes other than the final loan payment, annual income level, and credit-related variables (changes in credit card limits, utilization rates, credit grades, and debt-to-income ratios). The residual, ϵ_{it} , captures changes in expenditure not explained by the final payment or the controls.

Since loan durations range from 36 to 60 months and the sample spans 20 quarters, individuals complete their loans at different points in the sample period, generating plausibly exogenous variation in the timing of the anticipated income gain. Our identification strategy exploits this cross-sectional variation in loan completion dates: because each individual's final payment date is determined solely by their own origination date and contract duration, it is independent of every other individual's final payment date and uncorrelated with unobserved determinants of spending.⁶ Treated individuals are those completing their final payment in a given quarter, and the control group comprises individuals with ongoing repayments throughout the observation window. We include three quarters of leads to verify that treated and control individuals follow parallel consumption trajectories prior to loan completion, and normalize all event-study estimates to the quarter immediately preceding completion as the baseline.

⁵The baseline specification includes one lag and three lead terms. Section 4.4 reports results under an extended window with two lag terms.

⁶This identification strategy follows Agarwal, Liu and Souleles (2007); Scholnick (2013); Kueng (2018), who exploit similar variation in the timing of predictable income changes to estimate consumption responses.

3.2 Identifying the Magnitude Effect

3.2.1 Size Classifications

We consider three classifications of the magnitude of the anticipated income change: (i) the absolute dollar amount of the final payment (FP), (ii) the payment size relative to the individual's quarterly before-tax income (FP to $Income$), and (iii) the payment size relative to the individual's typical quarterly consumption expenditure in the two quarters preceding loan completion (FP to CCE). Absolute size captures the nominal stakes of the income change and provides the most direct comparator to the fiscal stimulus literature, where transfers are typically described in dollar terms. Relative size captures the economic salience of the change: a \$500 payment represents a meaningful income gain for a household earning \$5,000 per quarter but is near-trivial for one earning \$50,000, even though the nominal amount is identical. Whether households perceive and respond to an income change is more plausibly governed by its size relative to their resource base than by its absolute value, which motivates our preference for size relative to income as the primary measure.

3.2.2 MPC Heterogeneity

A primary objective of this analysis is to evaluate how consumption responses vary across the distribution of payment sizes and individual observable characteristics. To investigate MPC heterogeneity, we categorize individuals into three subgroups based on the distribution of absolute and relative payment sizes: Low (bottom 25th percentile; the reference group), Middle (25th–75th percentile), and High (top 25th percentile).⁷ This classification captures both cross-sectional variation across different payment types and idiosyncratic variation within specific size distributions. Furthermore, we examine the roles of age, income, and liquidity to determine the marginal impact of each variable conditional on others.

MPC Heterogeneity by Size Distribution.— To estimate the heterogeneity of consumption responses across size groups, we utilize an indicator function, $\mathbb{1}(y_{it} \in D)$, within the following parametric regression framework:

$$\Delta c_{it} = \alpha_t + \gamma_i + Region_i + \sum_D \beta_D \cdot FP_{it} \times \mathbb{1}(y_{it} \in D) + \lambda' x_{it} + \epsilon_{it} \quad (2)$$

⁷In Section 4.4, we use an alternative grouping strategy based on five quintiles as a robustness check.

where $y_{it} \in \{FP, FP \text{ to Income}, FP \text{ to CCE}\}$ represents the variable of interest, and $D \in \{Low, Middle, High\}$ denotes the distributional group. The coefficient β_D identifies the change in consumption for each size-based group D relative to the payment y_{it} . We estimate these effects using both absolute payment size and relative payment size scaled by income and consumption.

Conditional MPC Heterogeneity.— We examine MPC variation based on payment size, conditioning on three key factors identified in existing literature: age, income, and liquidity. The specification for the conditional consumption response is:

$$\Delta c_{it} = \alpha_t + \gamma_i + Region_i + \sum_{D_z} \beta_{D_z} \cdot FP_{it} \times \mathbb{1}(z_{it} \in D_z) + \sum \delta_{D_z} \times \mathbb{1}(z_{it} \in D_z) + \lambda' x_{it} + \epsilon_{it} \quad (3)$$

where $z_{it} \in \{Age, Income, Liquidity\}$ represents the observable factor for each tercile D_z . Our primary parameter of interest, β_{D_z} , assesses whether these characteristics systematically shift the consumption response to the auto loan repayment (FP_{it}).

To implement this, we stratify the sample into 3×3 subgroups by crossing the payment size with each factor tercile. This joint distribution allows for a direct interpretation of how MPCs vary across these dimensions. In the absence of direct wealth data, we follow [Fuchs-Schündeln and Hassan \(2016\)](#) by using quarterly income and mortgage debt status as proxies for liquidity constraints.⁸ Specifically, we define individuals as liquidity-constrained if they exhibit low income or carry additional mortgage debt alongside their auto loan.

4 Effects of Anticipated Income Changes

4.1 Baseline Consumption Response

Table 2 reports the baseline consumption response to the predictable income change at loan completion. Across the four specifications, the estimated MPC ranges from 0.177 to 0.196. Our preferred specification (Column 4), which includes individual, time, and region fixed effects along with the full set of controls, yields an MPC of 0.177, statistically significant at the one percent level and within the range documented in comparable studies (0.12 to 0.43; see Table A.1). The results

⁸While some research discusses wealthy hand-to-mouth individuals who hold substantial illiquid assets but have limited liquid funds, we assume income remains a reasonable proxy for liquidity constraints, as this group behaves similarly to “poor hand-to-mouth” individuals ([Kaplan, Violante and Weidner, 2014](#)).

Table 2: Consumption Response to Anticipated Income Changes

Dep. Var: Δc_{it}	(1)	(2)	(3)	(4)
FP	0.190*** (0.032)	0.178*** (0.032)	0.196*** (0.034)	0.177*** (0.033)
Constant	0.237 (0.152)	0.219 (0.156)	0.266 (0.167)	0.393* (0.218)
Control Variables	No	Yes	No	Yes
Time and Region FE	Yes	Yes	Yes	Yes
Individual FE	No	No	Yes	Yes
R-squared	0.003	0.028	0.003	0.059
Observations	77,148	77,148	77,148	77,148

Notes: FP indicates the final car loan payment level. Control variables include the changes in income, annual income level, the changes in credit card limits, credit card utilization rates, credit grades, debt to income ratios, and age dummies (30-39, 40-49, 50-59, 60-69, and 70+). Considering the measurement errors, observations with final payments greater than 1.5 are excluded from the sample. Robust standard errors in parentheses are clustered at the individual level. *, **, *** represent the significance level at 10%, 5%, and 1%, respectively.

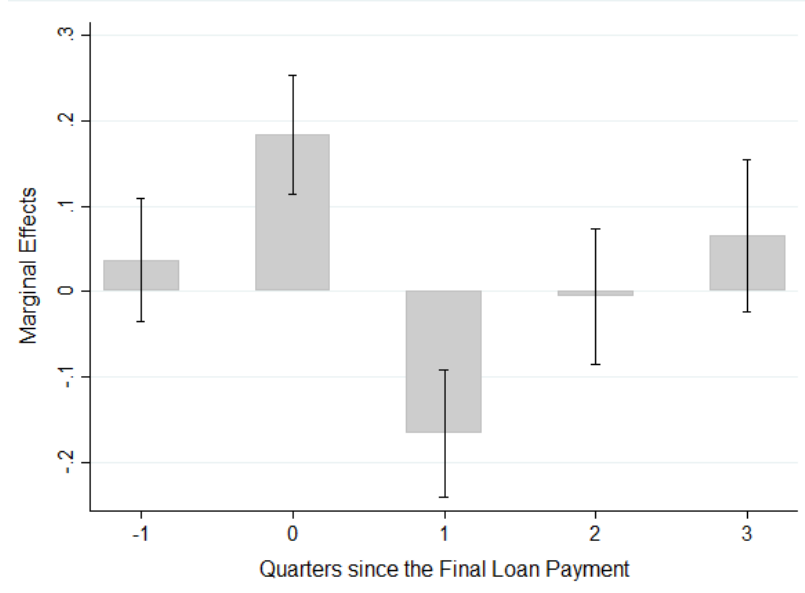
are robust to alternative normalizations of payment size: Table D.1 reports qualitatively similar estimates of excess sensitivity when the final payment is scaled by income and by consumption expenditure.

Figure 3 examines the temporal profile of consumption responses. Panel (a) plots the marginal effects β_s in equation (1) over horizons $s \in [-1, +3]$. Two features are noteworthy. First, the pre-treatment estimate at $s = -1$ is small and statistically insignificant at the 95 percent confidence level, providing evidence that individuals do not adjust spending in anticipation of the income gain. Second, the response at $s = 0$ is large and statistically significant (point estimate: 0.18), and declines sharply toward zero in subsequent quarters. Panel (b) confirms this pattern: the cumulative MPC effects are 0.04, 0.22, 0.05, 0.04, and 0.11 for periods $t - 1$ through $t + 3$, respectively, peaking at loan completion and reverting toward zero within two quarters thereafter.⁹ Taken together, the sharp one-time spike at realization combined with the absence of pre-completion smoothing is jointly inconsistent with the permanent income hypothesis under full rationality, irrespective of whether liquidity constraints bind.

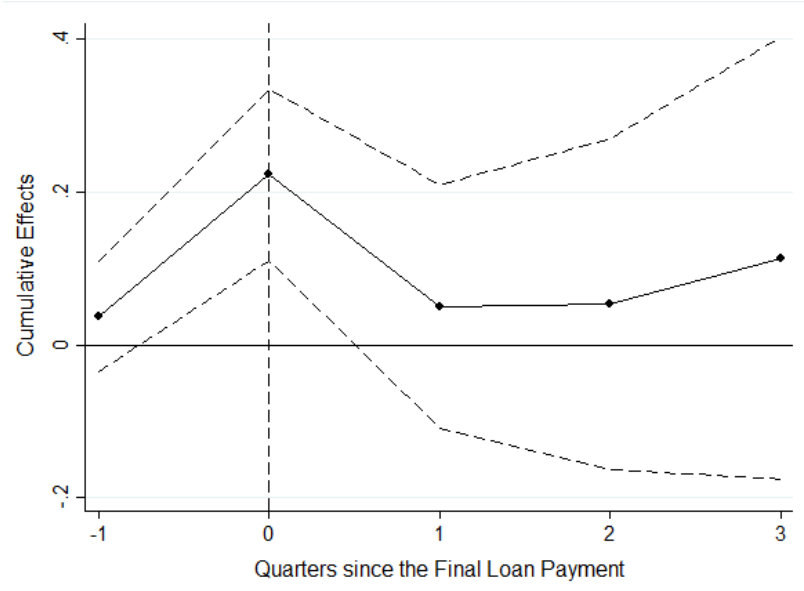
⁹Appendix C documents the corresponding income dynamics. Figure C.1 shows that income stabilizes at a persistently higher level following the final loan payment, consistent with the interpretation that borrowers treat the payment relief as a permanent income gain.

Figure 3: Consumption Response by Time

(a) Marginal Effects on Marginal Propensity to Consume



(b) Cumulative Effects on Marginal Propensity to Consume



Notes: Panel (a) plots the leads and lags of the marginal consumption response estimated from equation (1), with bars denoting point estimates and lines denoting 95 percent confidence intervals. Panel (b) plots the corresponding cumulative effects following the final car loan payment. The horizontal axis measures quarters relative to the quarter of loan completion. Standard errors are clustered at the individual level.

4.2 The Magnitude Effect

The Magnitude Effect on Excess Sensitivity.— Table 3 presents MPC estimates by tercile of each size measure. For absolute payment size (Column 1), the reference group (bottom 25 percent, < \$400 per quarter) exhibits an MPC of 0.758, approximately 76 cents of additional spending per dollar of income gain. This falls monotonically to 0.20 for the middle tercile and 0.14 for the top tercile. For payment size relative to quarterly income (Column 2), the gradient is even steeper: the reference group MPC is 0.712 and the top-tercile MPC falls to 0.147, a decline of over 0.56. Scaling by consumption expenditure (Column 3) yields qualitatively similar but somewhat attenuated results. In all cases, the decline in MPC with payment size is monotonic and statistically significant.

The multivariate analysis in Columns (4) through (6) assesses which size measure most powerfully explains the variation in excess sensitivity. When absolute size and the payment-to-income ratio are entered together (Column 4), the relative-size interaction terms retain significance while the contribution of absolute size diminishes considerably. Columns (5) and (6) corroborate this pattern: the payment-to-income ratio consistently dominates both absolute size and the consumption-scaled measure. This suggests that while the absolute value of the income gain matters, its effect on excess sensitivity is amplified when measured relative to the household’s income base, indicating that economic salience relative to income is the more fundamental driver.

Payment Size and Quarterly Income.— A potential concern is spurious correlation between payment size and income, as households may systematically select vehicles and down payment sizes commensurate with their income level. We address this concern in two ways. First, plotting absolute payment size and the payment-to-income ratio against quarterly income levels (Figure E.1) reveals that payment size is largely invariant across the income distribution in both absolute and relative terms, with a weak correlation coefficient of 0.20. Within each payment size tercile, borrowers span a broad range of income levels, suggesting that payment size reflects loan terms rather than income. Second, all specifications control directly for annual income levels, income changes other than the final payment, and debt-to-income ratios, so the interaction terms isolate variation in payment magnitude holding income constant. Together, these confirm that the heterogeneity across payment size reflects genuine variation in the magnitude of the income gain rather than differences in household income.

Table 3: The Effect on Consumption by Absolute and Relative Magnitudes

Dep. Var: Δc_{it}	(1)	(2)	(3)	(4)	(5)	(6)
FP (reference group)	0.758*** (0.156)	0.712*** (0.158)	0.321*** (0.066)	0.863*** (0.169)	0.761*** (0.156)	0.712*** (0.158)
FP * $\mathbb{1}$ (FP=Middle)	-0.558*** (0.164)			-0.308 (0.218)	-0.502*** (0.170)	
FP * $\mathbb{1}$ (FP=High)	-0.614*** (0.160)			-0.343 (0.229)	-0.492*** (0.182)	
FP * $\mathbb{1}$ (FP to Income=Middle)		-0.540*** (0.165)		-0.378* (0.217)		-0.474*** (0.173)
FP * $\mathbb{1}$ (FP to Income=High)		-0.565*** (0.163)		-0.378* (0.228)		-0.417** (0.192)
FP* $\mathbb{1}$ (FP to CCE=Middle)			-0.184** (0.075)		-0.129 (0.092)	-0.144 (0.100)
FP* $\mathbb{1}$ (FP to CCE=High)			-0.225 (0.153)		-0.172 (0.169)	-0.199 (0.177)
Constant	0.390* (0.218)	0.396* (0.218)	0.393* (0.218)	0.393* (0.218)	0.392* (0.218)	0.396* (0.218)
R-squared	0.059	0.059	0.059	0.059	0.059	0.059
N	77,148	77,148	77,148	77,148	77,148	77,148

Notes: FP, FP to Income, and FP to CCE indicate the absolute size of final car loan payment, final payment to quarterly before-tax income ratio, and final payment to quarterly consumption expenditure ratio, respectively. The reference group is defined as the bottom 25 percent of size distribution. Control variables include the changes in income, annual income level, the changes in credit card limits, credit card utilization rates, credit grades, debt to income ratios, and age dummies (30-39, 40-49, 50-59, 60-69, and 70+). Considering the measurement errors, observations with final payments greater than 1.5 are excluded from the sample. Robust standard errors in parentheses are clustered at the individual level. *, **, *** represent the significance level at 10%, 5%, and 1%, respectively.

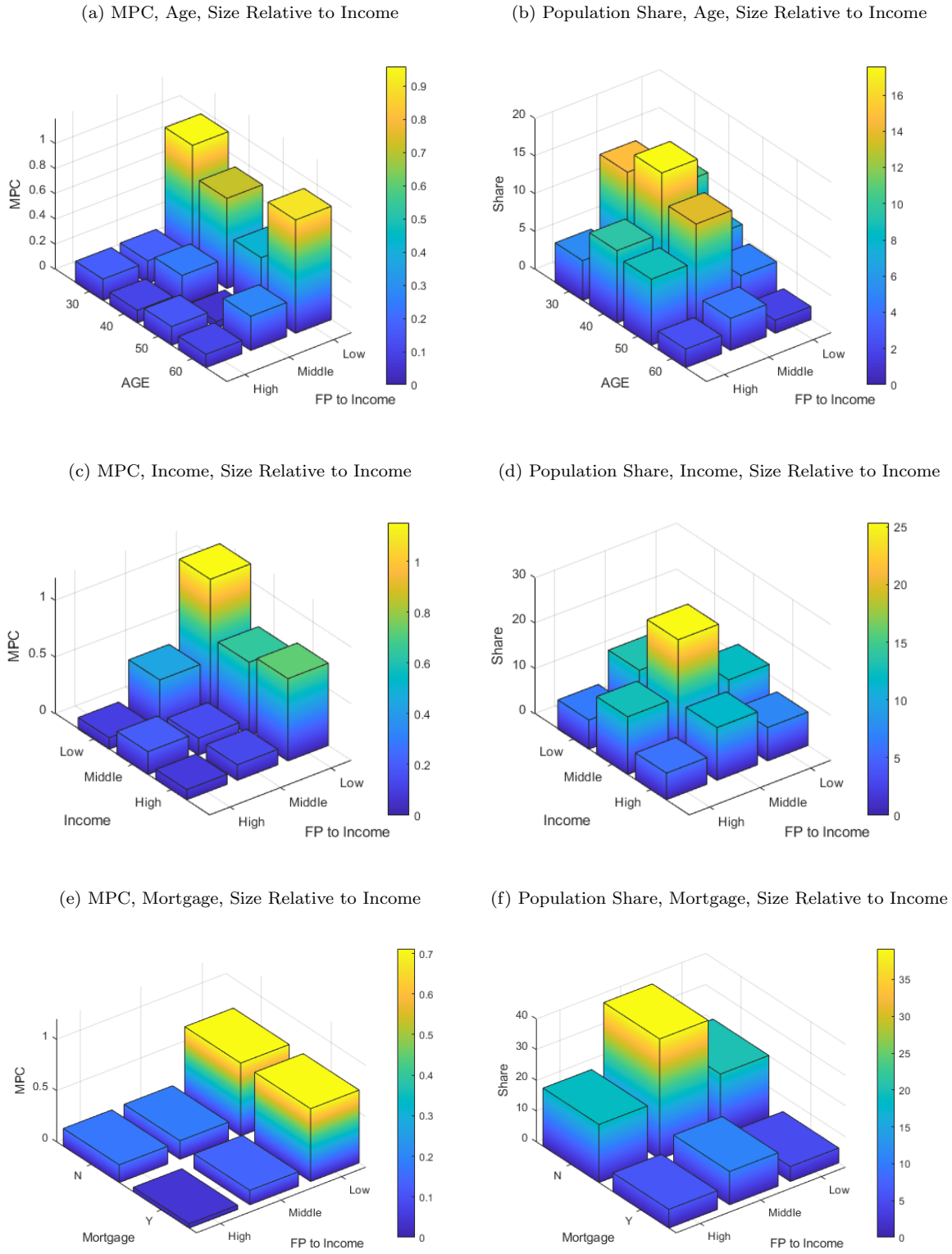
4.3 Conditional MPC Heterogeneity

Our main estimates establish that payment-size magnitude is a powerful predictor of excess sensitivity. A central question is whether this magnitude effect operates as an independent dimension of heterogeneity or merely proxies for factors commonly emphasized in the literature, such as age, income, and liquidity constraints. We focus on the payment-to-income ratio as the primary size measure, given that it is the most significant determinant among the three normalizations considered in the previous section.¹⁰

Age.— Figure 4, Panel (a), plots the three-by-three MPC grid across age groups and payment-size terciles. The MPC is highest in the low-size group and declines monotonically as payment size increases across every age cohort. Age exerts no systematic effect on the level of the MPC once

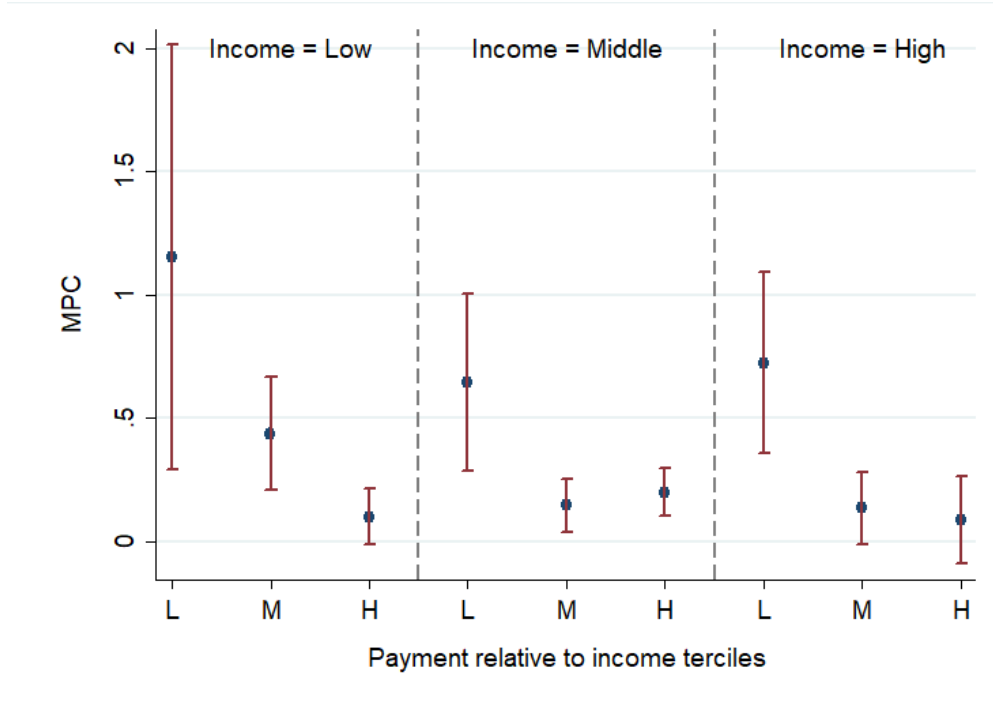
¹⁰Figure F.1 reports conditional MPC heterogeneity across absolute payment size, age, and income.

Figure 4: Heterogeneous consumption responses



Notes: Left panels plot conditional MPC estimates; right panels plot population shares across payment-to-income terciles, by age (a,b), income (c,d), and mortgage status (e,f). Bar color reflects the magnitude of the MPC or population share, as indicated by the respective color scales.

Figure 5: MPC by income given the relative size to income (FP to Income)



Notes: This figure plots MPC estimates across payment-to-income terciles (L = low, M = middle, H = high) within each income group. Dots denote MPC point estimates and bars denote 95 percent confidence intervals with standard errors clustered at the individual level.

payment size is held constant: the bars within a given size tercile are broadly similar in height across age groups. Panel (b) documents that observations are concentrated in the 40 to 50 age range, where the patterns are therefore most precisely identified.

Income.— Figure 4, Panel (c) presents the decomposition across income terciles. Within each income group, the MPC declines steeply with payment size, replicating the aggregate pattern. Across income groups at a fixed payment size, lower-income individuals exhibit somewhat higher MPCs, consistent with the conventional liquidity-constraint narrative. However, the within-income variation driven by payment size substantially exceeds the between-income variation at a given size: moving from the lowest to the highest payment size tercile within the low-income group reduces the MPC by approximately 0.65, whereas moving from low to high income within the lowest size tercile reduces it by only 0.25.

Figure 5 confirms this pattern more precisely, plotting the MPC by relative size tercile separately for each income group with 95 percent confidence bands. Within every income group, the MPC

declines monotonically from the lowest to the highest payment-size tercile, and the gap between the two extremes is statistically significant in each case. Strikingly, high-income households facing a low payment-to-income ratio exhibit MPCs comparable to those of low-income households in the same size tercile, whereas low-income households facing a high payment-to-income ratio display responses as muted as their high-income counterparts. A formal test comparing high-income individuals in the low-size group against low-income individuals in the high-size group yields an F-statistic of 7.11 ($p < 0.01$), confirming that the relative magnitude of payment relief dominates income level as the primary predictor of excess sensitivity.¹¹

Liquidity.— To further probe the liquidity mechanism, we exploit variation in mortgage debt status as an additional proxy for binding borrowing constraints (Panel (e) of Figure 4).¹² Despite this, the magnitude effect is equally pronounced for both groups: individuals with small relative payment sizes exhibit high MPCs regardless of mortgage status, and the MPC profile declines similarly across the payment-size distribution for both the mortgaged and non-mortgaged subsamples. This pattern is difficult to reconcile with a pure liquidity story.

An important caveat is that mortgage status and income are imperfect proxies for liquidity, as [Kaplan and Violante \(2014\)](#) emphasize. We cannot fully rule out the wealthy hand-to-mouth channel without direct wealth data. However, the fact that the magnitude effect is equally strong in both high-income and low-income groups, and in both mortgaged and non-mortgaged groups, suggests that it is not simply capturing a subset of constrained households. These results indicate that the magnitude of the anticipated income change is an independent predictor of excess sensitivity that cannot be attributed to observationally measurable liquidity differences.

4.4 Robustness Checks

We conduct three robustness checks to validate our main results. First, we test whether the monotone decline in the MPC is sensitive to the choice of grouping strategy. Second, we examine heterogeneity in consumption dynamics beyond the average trajectory. Lastly, we re-estimate the baseline results in Korean won using an extended analysis window and alternative specifications.

¹¹Figure F.2 conditions on payment-size tercile and finds MPCs statistically indistinguishable across income terciles within each group, ruling out income level as an independent driver of excess sensitivity.

¹²We extend our analysis to other liquidity-related factors such as high credit utilization, credit card consolidation loans, late payments, unused credit lines, and default risk, but the limited sample sizes for these subgroups preclude robust statistical inference.

Alternative Grouping Strategy.— Our baseline analysis classifies payment size into three subgroups. To assess sensitivity to this choice, we extend the analysis to five quintiles following Kueng (2018), estimating

$$\Delta c_{it} = \alpha_t + \gamma_i + Region_i + \sum_{q_y} \beta_{q_y} \cdot FP_{it} \times \mathbb{1}(y_{it} \in q_y) + \sum_{q_y} \gamma_{q_y} \times \mathbb{1}(y_{it} \in q_y) + \lambda' x_{it} + \epsilon_{it}, \quad (4)$$

where y_{it} is the payment-to-income ratio and $\mathbb{1}(y_{it} \in q_y)$ indicates whether individual i falls within quintile q , with $q1$ denoting the lowest and $q5$ the highest 20 percent of the distribution. The coefficient, β_{q_y} , measures the consumption response to a one-unit increase in the payment-to-income ratio within quintile q , and the term γ_{q_y} captures quintile-specific intercepts that absorb mean differences in consumption growth across payment size groups, ensuring that β_{q_y} isolates the consumption response to the income gain within each quintile.

Appendix G, Figure G.1 plots the β_{q_y} estimates across quintiles. The monotone decline in the MPC is preserved under this finer grouping: point estimates fall from 0.85 in the lowest quintile to 0.11 in the highest, and the gradient remains statistically significant throughout, with non-overlapping confidence intervals across adjacent quintiles at the tails of the distribution.¹³ Individuals for whom the payment represents a small share of quarterly income spend nearly all of the predicted income gain, while those with large relative payments smooth consumption more optimally, exhibiting substantially lower excess sensitivity.

Consumption Dynamics by Size and Income Group.— We replicate the baseline estimation separately across terciles defined by the payment-to-income ratio and by income level. Figures G.2 and G.3 report the marginal and cumulative MPC estimates for the low, middle, and high payment-size groups. The cumulative dynamics broadly mirror the aggregate pattern documented in Figure 3: the consumption response peaks at loan-completion (i.e., $s = 0$) and reverts toward zero within two quarters across all size groups. The magnitude of the response, however, is substantially attenuated for the middle and high payment-size terciles.

Figure G.4 replicates this exercise across income terciles. Low- and middle-income households display qualitatively similar dynamics to the baseline, with cumulative MPCs peaking at $s = 0$ before reverting. The high-income tercile, by contrast, exhibits statistically insignificant consump-

¹³The point estimates of the spending response across payment-to-income ratio quintiles, from the lowest to the highest, are 0.85, 0.35, 0.20, 0.12, and 0.11, respectively.

tion responses at all horizons, consistent with near-optimal consumption smoothing. This pattern accords with theory: for high-income households, the payment relief constitutes a smaller share of the income base, limiting the scope for excess sensitivity. Together, the heterogeneity across both dimensions suggests that it is the relative magnitude of the predictable income change rather than the level of income per se that governs the extent of excess sensitivity.

Currency and Specification Robustness.— Converting Korean won to U.S. dollars using the sample-period mean exchange rate may introduce systematic bias, potentially confounding comparisons with the baseline estimates. To address this, we replicate the baseline estimation in the original currency. As reported in Table G.1 the excess sensitivity estimate is unchanged at 0.177, confirming that the currency conversion does not materially affect the results.

We also assess robustness to alternative specifications of the dependent variable. Replacing the level change in consumption expenditure with the log difference yields an estimated elasticity of 0.35: a one-percent anticipated income increase is associated with a 0.35 percent rise in consumption growth. Finally, we double the sample size and extend the event window to two quarterly lags and four quarterly leads to evaluate the persistence of the estimated effects (See Figure G.5). The marginal consumption response consistently peaks in the quarter of loan completion across all specifications, reinforcing the robustness of the baseline findings.

Taken together, the evidence in this section establishes three consistent findings. First, the average MPC of approximately 0.18 confirms significant excess sensitivity to a fully anticipated income gain, consistent with prior estimates and at odds with the LCPIH. Second, the temporal profile sharpens this inference: spending rises discretely at realization with no anticipatory response and mean reversion within two quarters, ruling out forward-looking smoothing irrespective of whether liquidity constraints bind. Third, and most importantly, the MPC declines monotonically from 0.76 to 0.14 across payment size terciles, a gradient that steepens when size is measured relative to income and where the relative measure retains independent predictive power over the absolute one in joint specifications. This gradient is not a compositional artifact: it persists within every age, income, and liquidity subgroup, and moving across payment size terciles within the low income group reduces the MPC by 0.65, whereas moving across income terciles at a fixed payment size reduces it by only 0.25. The relative economic magnitude of the anticipated income change is thus the dominant determinant of excess sensitivity, a dimension of heterogeneity that liquidity constraints and rule-of-thumb behavior alone cannot accommodate.

5 Potential Explanations for MPC Heterogeneity

Our empirical results are jointly inconsistent with the standard LCPIH and with several of its most common extensions. This section works through the leading candidate mechanisms and argues that bounded rationality, in which agents selectively optimize their consumption path depending on the stakes involved, provides the most coherent and parsimonious account of the evidence. We then quantify the welfare cost of failing to smooth consumption, which provides a micro-foundation for why the degree of deviation from optimal behavior depends so systematically on the size of the income change.

5.1 Standard Models and Their Limitations

A fully rational forward-looking agent should exhibit near-zero MPCs for predictable income changes of any size, since the income gain should already be incorporated into the consumption path well before realization. Our finding of a significant average MPC of 0.18, together with the absence of any pre-event smoothing, rules out this benchmark.

Liquidity constraints are the natural first alternative: a household unable to borrow against future income must defer consumption until cash arrives, generating a high MPC at realization (Carroll, 1997; Kaplan and Violante, 2014). This channel cannot, however, account for the magnitude effect. A constrained household should respond strongly to income changes of all sizes, yet we find substantially lower MPCs for large income changes even within low-income and high-mortgage-debt subgroups. The post-event consumption profile compounds the difficulty: a relaxing liquidity constraint should produce a step up to a permanently higher consumption level, whereas we observe a transient spike that reverts to baseline within two quarters. While low-income households do exhibit larger spending adjustments within any given payment-size group, consistent with the broader literature (Garcia, Lusardi and Ng, 1997; Johnson, Parker and Souleles, 2006; Parker, 2017; Coibion, Gorodnichenko and Weber, 2020), this effect holds only conditional on payment size and is substantially weaker than the size gradient itself.

Two further alternatives also fall short. Rule-of-thumb behavior generates a constant MPC of approximately one irrespective of income-change size (Campbell and Hercowitz, 2019), which is at odds with the monotone gradient from 0.85 to 0.11 we document. Myopia offers a partial account in that individuals who perceive freed-up loan payments as transitory rather than permanent may allocate them toward immediate spending, but it predicts uniformly high MPCs regardless of

magnitude and is silent on why the response concentrates in a single quarter with no anticipatory adjustment. Risk aversion and life-cycle effects may also contribute to deviations from full smoothing (Meghir and Pistaferri, 2011; Pagel, 2017), yet neither generates a prediction that the degree of deviation should decline monotonically with income-change size. A mechanism that can simultaneously account for the level, the gradient, and the timing of the spending response is therefore needed, which we turn to next.

5.2 Bounded Rationality and Selective Optimization

Bounded rationality provides a more coherent account of the evidence. The core idea, developed in the context of consumption by Browning and Collado (2001) and Hsieh (2003), is that agents do not uniformly apply intertemporal optimization to every income change they face. Instead, they selectively devote cognitive resources to recalculating their consumption path when the welfare gain from doing so is sufficiently large. For small income changes, the utility loss from suboptimal adjustment is modest, so households have little incentive to plan ahead and instead spend the cash when it arrives, generating high excess sensitivity. Conversely, for large income changes, the high cost of suboptimality motivates consumers to plan ahead, resulting in lower MPCs that align more closely with traditional consumption theory.

This mechanism is consistent with the consumption responses we document. The concentration of the spending response at realization, with no pre-event adjustment even for agents who formally anticipate the income change, is consistent with agents for whom the expected utility gain from acting on that anticipation falls below the cognitive threshold. The monotone decline in MPCs as size increases reflects progressively more agents finding optimization worthwhile. The stronger predictive power of relative size over absolute dollars also accords with this view, since the welfare gain from optimal smoothing scales with the income change as a share of the household’s permanent income rather than its nominal value. These findings connect to a growing body of evidence on size-dependent consumption responses. Andre et al. (2025) document that stated MPCs from hypothetical income changes decline with income-change size in survey experiments, and that respondents report lower planning effort for smaller windfalls. Baugh et al. (2021) and Graham and McDowall (2026) document related asymmetries in consumption smoothing attributable to inattention and mental accounting.

5.2.1 Welfare Cost Analysis

To further support the bounded rationality interpretation, we quantify the welfare costs of deviating from the optimal consumption path. Following [Fuchs-Schündeln and Hassan \(2016\)](#) and [Kueng \(2018\)](#), we adopt a sufficient statistics approach in which the welfare loss from failing to fully smooth consumption is approximated by the squared deviation between the actual and optimal consumption paths, weighted by the marginal utility of consumption. Formally, the potential loss is given by

$$Welfare\ loss\ (c_i^{deviate}, c_i^{pih}) \approx \frac{\delta}{2} \cdot \sum_t \zeta_t \left(\frac{c_t^{deviate} - c_t^{pih}}{c_t^{pih}} \right)^2, \quad (5)$$

where δ captures the curvature of the utility function, and ζ_t is the utility weight function at period t . We assume a constant relative risk aversion utility function of the form $u(c) = c^{1-\delta}/(1-\delta)$, and set $\delta = 2$, the standard value in the literature.¹⁴ Applying the envelope theorem, equation (5) simplifies to

$$Welfare\ loss\ (c_i^{deviate}, c_i^{pih}) \approx \frac{\delta}{2} \cdot \left[(1 - MPC) \cdot FP_i / c_i^{pih} \right]^2,$$

where FP_i / c_i^{pih} denotes the final car loan payment scaled by the individual's average consumption, which serves as a proxy for permanent income. The full derivation is provided in Appendix [H](#).

Substituting the estimated MPCs and mean payment sizes for each tercile yields welfare losses of 0.13, 0.61, and 2.4 percent of permanent income for the low, middle, and high payment-size groups, respectively. The welfare cost rises monotonically with the relative size of the income change. For households in the bottom tercile, the utility cost of failing to smooth is negligible at less than 0.15 percent of permanent income, providing little incentive to incur the cognitive costs of reoptimization. For households in the top tercile, the cost reaches 2.40 percent of permanent income, a magnitude sufficient to render deliberate planning worthwhile. The monotone welfare cost profile thus offers a coherent micro-foundation for the bounded rationality interpretation: rather than reflecting a structural departure from optimizing behavior, the observed decline in excess sensitivity with payment size is consistent with a rational allocation of cognitive resources in which the returns to careful planning increase with the stakes.

¹⁴Under this assumption, the utility weight function satisfies $\zeta_t = \gamma^t \frac{\partial u(c_t^{pih})}{\partial c} c_t^{pih} / \sum_i \gamma^n \frac{\partial u(c_n^{pih})}{\partial c} c_n^{pih} = \frac{\gamma^t u(c_t^{pih})}{U(c^{pih})}$, where γ is the discount factor.

6 Policy Implications

6.1 From Micro Evidence to Fiscal Design

The magnitude effect raises a question that is relevant, though not straightforwardly answered, by our setting: does the size of an individual transfer, and not just its recipient, independently shape the aggregate spending response to fiscal policy? The conventional targeting argument holds that concentrating transfers on low-income or liquidity-constrained households maximizes the fiscal multiplier because such households tend to exhibit the highest MPCs (Jappelli and Pistaferri, 2014; Coibion, Gorodnichenko and Weber, 2020). Our evidence introduces an additional margin: within any recipient group, the amount each household receives independently determines its spending response, since the MPC is decreasing in transfer size relative to income. We illustrate the importance of this margin through a policy experiment that holds total outlays fixed while varying the distribution of payments, following the approach of Jappelli and Pistaferri (2014).

Our setting shares three features with fiscal stimulus that support the comparison. Both auto loan completions and stimulus payments are either predetermined or announced in advance, giving recipients prior knowledge of the size and timing of the income gain. This satisfies the foreknowledge assumption standard in the fiscal multiplier literature, where implementation lags between announcement and receipt are well documented. In addition, both represent irregular rather than recurring income changes, distinguishing them from predictable annual flows such as tax refunds. Fuchs-Schündeln and Hassan (2016) show that this distinction matters empirically: households tend to smooth consumption in response to regular, repeated income changes but exhibit significant spending responses at receipt for irregular transfers, precisely the category into which both auto loan completions and fiscal stimulus payments fall. Beyond these structural similarities, the magnitudes are also comparable: the historical range of US stimulus payments averages \$800 with individual payments between \$500 and \$1,200, closely in line with our payment range of \$421 to \$1,040, lending further support to the calibration.¹⁵

We nonetheless acknowledge important differences that bear on interpretation. Auto loan completion generates a permanent increase in disposable income while fiscal transfers are typically transitory. Since permanent income changes tend to elicit stronger smoothing responses, our MPCs

¹⁵The 2001 income tax rebates averaged \$500 per individual. The 2008 and 2020 economic stimulus payments averaged \$900 and \$1,200, respectively, targeting incomes below \$75,000.

likely constitute an upper bound for the spending response to transitory fiscal transfers, and direct comparisons should be made with this caveat in mind. Beyond the permanence distinction, the framing of the income source may independently shape behavior: evidence from the tax rebate literature suggests that how households mentally account for a windfall affects their propensity to spend it (Sahm, Shapiro and Slemrod, 2010). Finally, our sample skews toward middle and upper income households relative to the targets of most fiscal programmes, likely attenuating the role of liquidity constraints that would amplify spending responses among lower-income recipients. The experiment should therefore be read as an illustration of the qualitative direction of the magnitude effect rather than a precise quantitative forecast.

6.2 Policy Experiment Design and Results

Each policy sets total transfers at one percent of national disposable income, equivalent to \$3 million in our sample economy.¹⁶ Policy 1 mirrors conventional design by targeting the bottom tercile of total income distribution with equal payments of \$1,420 to recipients with annual income below \$28,150. Policy 2 extends coverage to the bottom two income terciles, distributing equal payment of \$470 to all households with income below \$40,800. Policy 2 therefore reaches approximately three times as many households at one-third the payment per recipient. Since the total outlays are identical by construction, any difference in aggregate outcomes isolates the effect of payment magnitude from the scale of fiscal intervention which is the margin we seek to examine.

For each policy, we evaluate outcomes under two MPC scenarios: a homogeneous MPC of 0.24, calibrated to the low-income tercile average, and the heterogeneous size-dependent MPCs estimated in Section 4. The aggregate MPC for policy $j \in \{1, 2\}$ is given by:

$$MPC_j = \sum_i MPC_i \times \frac{\Delta income_i(j)}{T} = \sum_i \beta_i \frac{\tau_i(j)}{T}, \quad (6)$$

where β_i is the estimated MPC for individual i and $\tau_i(j)$ is the transfer received by individual i under policy j .¹⁷ T denotes total government outlays, defined as $T = 0.01 \times \sum_i y_i$ where y_i is

¹⁶This is defined as the sum of each individual's disposable income in the final sample used in our main estimation.

¹⁷The transfer payment received by individual i in policy experiment j is equal to $\tau_i(j) = T/d^j \times \mathbf{1}(i \in t^j)$, where d^j is the total number of transfer recipients for policy j and $\mathbf{1}(i \in t^j)$ is an indicator function of the status of the transfer recipient.

Table 4: Effect of Government Transfers on Consumption Response

Policy Transfer: 1 percent of GDP	Aggregate MPC	Aggregate Consumption Growth
Homogeneous MPC		
Transfer to 1st bottom income tercile	0.24	0.45%
Heterogeneous MPC		
Transfer to 1st bottom income tercile	0.25	0.47%
Transfer to 1st and 2nd bottom income tercile	0.73	1.38%

Notes: Policy 1 targets the bottom income tercile, covering the lowest 25 percent of the income distribution. Policy 2 extends coverage to the bottom two income terciles, reaching the lowest 75 percent. Total transfers are held constant at one percent of national disposable income across all scenarios.

disposable income. Aggregate consumption growth under policy j is then computed as:

$$g(C)_j = \frac{\sum_i \beta_i \tau_i(j)}{\sum_i c_i} \quad (7)$$

where $g(C)_j$ denotes aggregate consumption growth for policy j , and c_i is the consumption expenditure for individual i .

Table 4 reports the results. Under the homogeneous MPC benchmark, Policy 1 generates an aggregate MPC of 0.24 and consumption growth of 0.45 percent. Introducing heterogeneous MPCs while holding the narrow targeting of Policy 1 fixed raises aggregate outcomes only marginally, to an MPC of 0.25 and consumption growth of 0.47 percent. The limited gain reflects a countervailing force: the large per recipient payment of \$1,420 places most recipients in the low MPC region of the size distribution, largely negating the benefit that MPC heterogeneity would otherwise provide. Heterogeneity alone, without a complementary adjustment to payment structure, yields negligible gains over the homogeneous benchmark.

Notably, Policy 2 under heterogeneous MPCs yields an aggregate MPC of 0.73 and consumption growth of 1.38 percent, nearly three times the benchmark at identical fiscal cost. The mechanism is straightforward: expanding coverage to a broader recipient pool reduces the payment-to-income ratio for each household, shifting recipients toward the high MPC region of the size distribution and amplifying the aggregate spending response. The resulting 0.91 percentage point gain in consumption growth relative to the Policy 1 is attributable entirely to this magnitude effect, and

would be absent under the assumption of a fixed MPC.

Taken together, these findings highlight a design margin that has received limited attention in the fiscal stimulus literature. Conditional on total outlays and distributional targeting, the per-recipient transfer amount independently shapes aggregate consumption responses by determining where recipients fall in the MPC distribution. Policies that distribute a given budget across more households through smaller payments may therefore dominate those that concentrate resources in fewer and larger transfers on purely stimulus grounds. This result complements recent work showing that the aggregate multiplier depends not only on who receives transfers but on the full distribution of household balance sheets and spending sensitivities ([Auclert, 2019](#); [Fagereng, Holm and Natvik, 2021](#)). Recognising the per recipient amount as a distinct policy instrument has quantitative implications for the aggregate multiplier that neither homogeneous MPC frameworks nor heterogeneous agent models that condition only on wealth and income are able to capture.

7 Concluding Remarks

This paper provides novel evidence on how the magnitude of an anticipated income change shapes household consumption responses. Exploiting the scheduled completion of fixed-term car loans as a natural experiment, we document a robust magnitude effect in which the MPC declines monotonically with the size of the income gain. This gradient steepens when payment size is expressed relative to household income and retains independent predictive power over absolute dollar amounts in joint specifications. The spending response concentrates sharply at the moment of income realization, with no anticipatory adjustment in the preceding period, a pattern inconsistent with forward-looking consumption smoothing irrespective of whether liquidity constraints bind.

Decomposing the MPC across age, income, and liquidity subgroups reveals that the magnitude effect is not a compositional artifact. Within every demographic and financial subgroup, MPCs decline steeply with payment size, and variation in payment size generates considerably more dispersion in MPCs than an equivalent shift across income groups at a fixed payment size. This asymmetry points to payment magnitude as an independent and quantitatively dominant predictor of excess sensitivity, one that observationally measurable liquidity differences alone cannot explain. We interpret this pattern through the lens of bounded rationality. When the welfare cost of failing to smooth is small, households have little incentive to devote cognitive resources to planning a response, generating high MPCs at realization. As payment size rises relative to income, the

cost of suboptimal adjustment grows sufficiently to motivate deliberate forward-looking behavior, yielding lower MPCs. A welfare cost analysis supports this interpretation, with the utility loss from failing to smooth rising monotonically across the payment-size distribution, providing a coherent and parsimonious micro-foundation for the observed gradient.

These findings carry direct implications for the design of fiscal transfer programs. A policy experiment holding total outlays fixed illustrates that distributing smaller payments across a broader recipient base substantially raises aggregate consumption growth relative to a conventional concentrated transfer. This gain is attributable entirely to the magnitude effect and would be absent under the assumption of a fixed MPC. The result points to a design margin that neither standard homogeneous MPC frameworks nor heterogeneous agent models that condition only on income and wealth are able to capture. The per-recipient transfer amount independently shapes the aggregate spending response by determining where recipients fall in the MPC distribution, and should be recognized as a distinct instrument of fiscal policy alongside the more commonly studied dimensions of targeting and total outlay.

More broadly, our results suggest that the magnitude of an income change is an essential consideration for both economic theory and policy design. Future research may extend this framework by examining magnitude effects in other transfer contexts, such as tax rebates and pension payouts, and by incorporating size-dependent MPCs into quantitative heterogeneous agent models to assess their aggregate implications more precisely.

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Appendix

A Literature Review

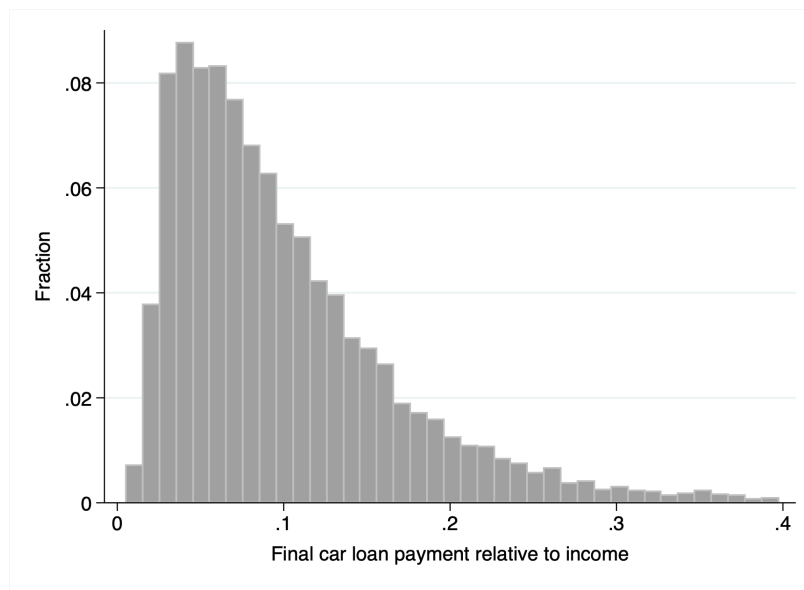
Table A.1: Overview of Marginal Propensity to Consume using Natural Experiments

Study	Experiment (USD)	Data	MPC (out of 1)	Liquidity constraint	Size
Agarwal et al. (2007)	2001 Federal income tax rebates (\$500)	Credit card accounts; 2000 - 2002	0.40	Based on credit limit, utilization rate, and age	No
Johnson et al. (2006)	2001 Federal income tax rebates (\$500)	CEX interview survey; 2000 - 2002	0.20 - 0.40	Based on age, income, and liquid assets	No
Misra and Surico (2014)	2001 Federal income tax rebates (\$500) & 2008 Economic stimulus payments (\$900)	CEX interview survey; 2000 - 2002 & 2007 - 2008	0.43 (2001) & 0.16 (2008)	Based on high income and high mortgage debt	No
Broda and Parker (2014)	2008 Economic stimulus payments (\$900)	Scanner data; 2007 - 2009	0.10	Availability of easily accessible funds	No
Parker et al. (2013)	2008 Economic stimulus payments (\$900)	CEX interview survey; 2007 - 2008	0.12 - 0.30	Based on age, income, and liquid assets	No
Scholnick (2013)	Last mortgage payment (\$627)	Credit card accounts; 2004 - 2006	0.40	Based on liquid assets	Yes
Kueng (2018)	Alaska permanent fund (\$1650)	Credit card accounts; 2010 - 2014	0.25	Based on income and liquid assets	Yes
Baker et al. (2020)	2020 Economic stimulus payments (\$1200)	Transaction level data; 2016-2020	0.25 - 0.40	Based on income and liquid assets	No
Coibion et al. (2020)	2021 Economic stimulus payments (\$1200)	Scanner data; 2018 - 2020	0.40	Based on income and liquid assets	No

Notes: This table reports the overview of MPC in response to an anticipated income increase based on natural experiments for each study. Each experiment has a corresponding amount in US dollars, which indicates the average amount received at an individual level. For 2020 Economic stimulus payments, we only list studies that examine the first-time payment made to households.

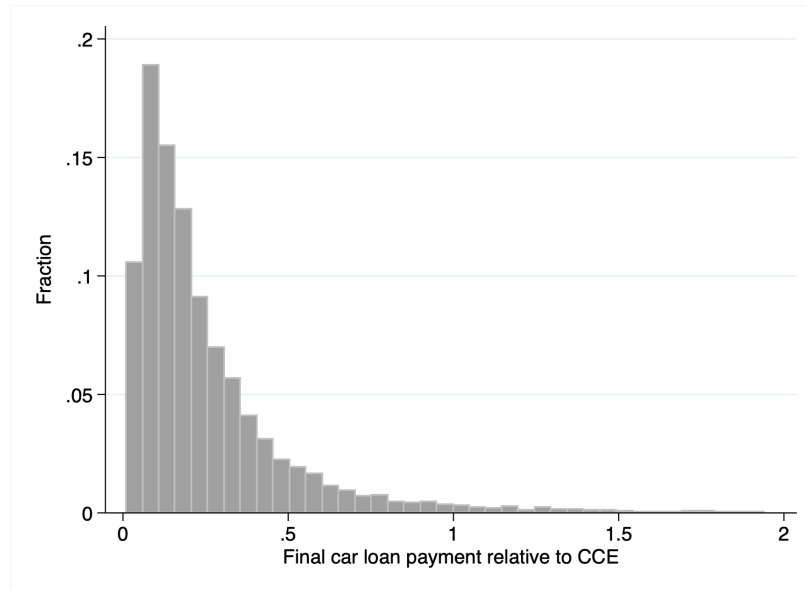
B Distribution of Sample

Figure B.1: Distribution of Payment Size Relative to Income, 2012-2016



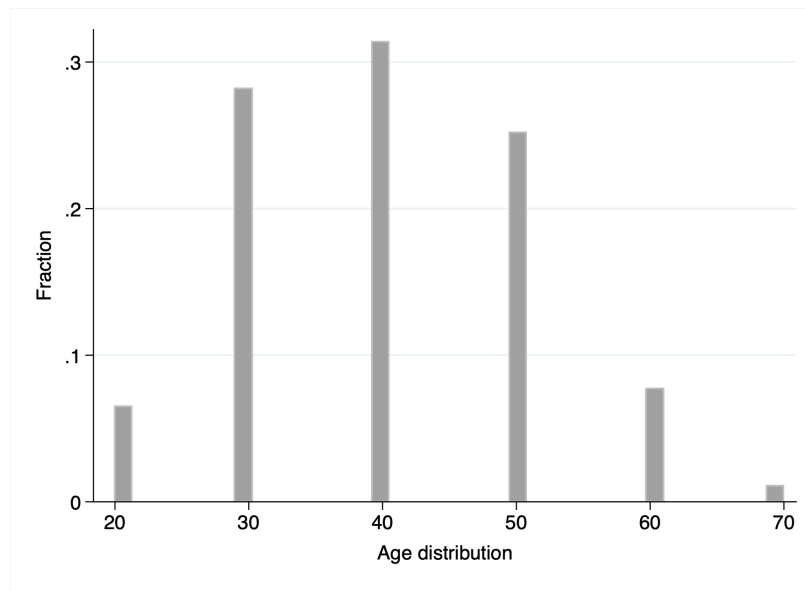
Notes: This figure plots the distribution of the final car loan payment relative to quarterly before-tax income for the estimation sample from 2012 to 2016. All values are CPI-adjusted to 2020 prices.

Figure B.2: Distribution of Payment Size Relative to Consumption, 2012-2016



Notes: This figure plots the distribution of the final car loan payment relative to two-quarter lagged credit card expenditure for the estimation sample from 2012 to 2016.

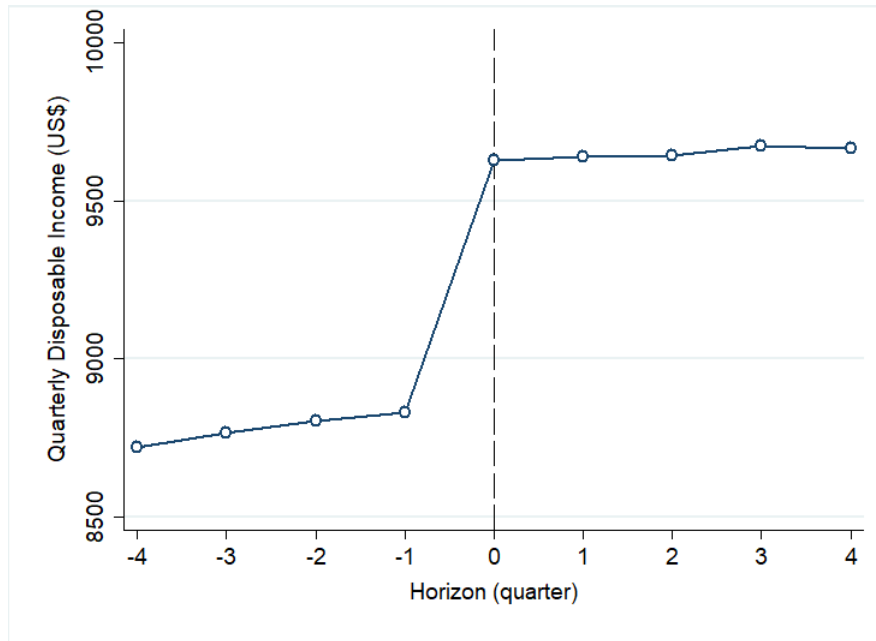
Figure B.3: Distribution of Age, 2012-2016



Notes: This figure plots the age distribution of the estimation sample from 2012 to 2016.

C Income Process

Figure C.1: Income Dynamics



Notes: This figure plots the quarterly income dynamics for final sample distribution. Dotted line indicates the event time ($t = 0$) where individuals have increase in income following the final car loan payment in $t - 1$ quarter.

D MPC by Relative Magnitudes

Table D.1: Consumption Response by Relative Magnitudes

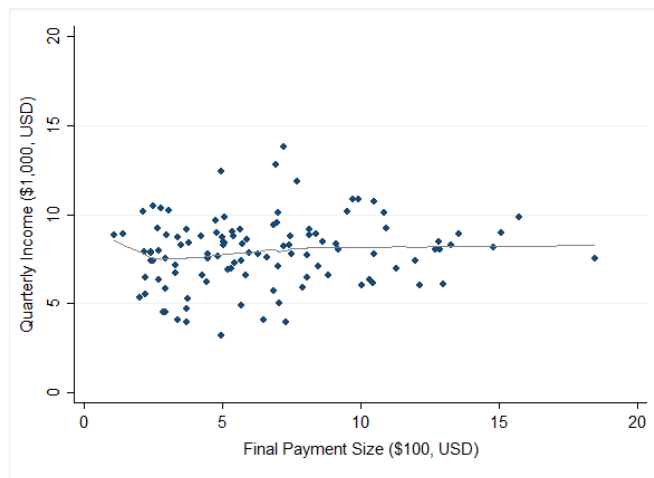
Dep. Var: Δc_{it}	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
FP to Income	1.609*** (0.243)	1.422*** (0.241)	1.626*** (0.255)	1.426*** (0.246)				
FP to CCE					0.690*** (0.053)	0.675*** (0.052)	0.658*** (0.057)	0.580*** (0.055)
Constant	-0.202 (0.146)	-0.216 (0.150)	-0.391** (0.166)	-0.224 (0.227)	-0.202 (0.146)	-0.205 (0.150)	-0.389** (0.166)	-0.216 (0.227)
Control Variables	No	Yes	No	Yes	No	Yes	No	Yes
Time, Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	No	No	Yes	Yes	No	No	Yes	Yes
R-squared	0.003	0.028	0.003	0.059	0.004	0.029	0.004	0.064
Observations	77,148	77,148	77,148	77,148	77,148	77,148	77,148	77,148

Notes: FP to Income and FP to CCE indicate the final payment size relative to income and consumption, respectively. Control variables include the changes in income, annual income level, the changes in credit card limits, credit card utilization rates, credit grades, debt to income ratios, and age dummies (30-39, 40-49, 50-59, 60-69, and 70+). Robust standard errors in parentheses are clustered at the individual level. *, **, *** represent the significance level at 10%, 5%, and 1%, respectively.

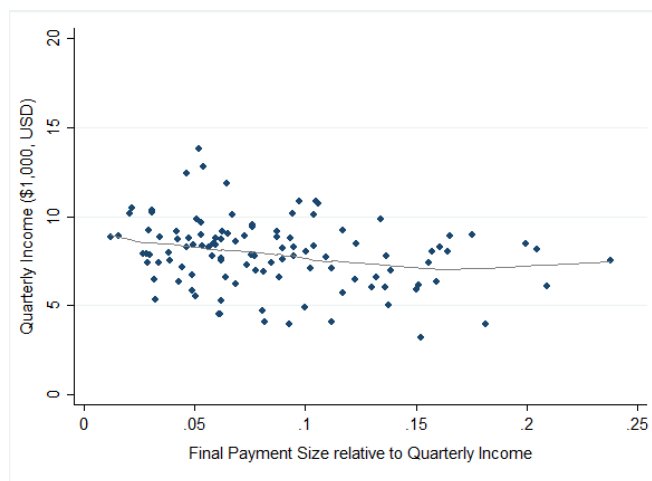
E Consumption Response by Relative Magnitudes

Figure E.1: Distribution of Final Car Loan Payment, 2012–2016

(a) Quarterly Income and Final Payment Size



(b) Quarterly Income and Relative Final Payment Size

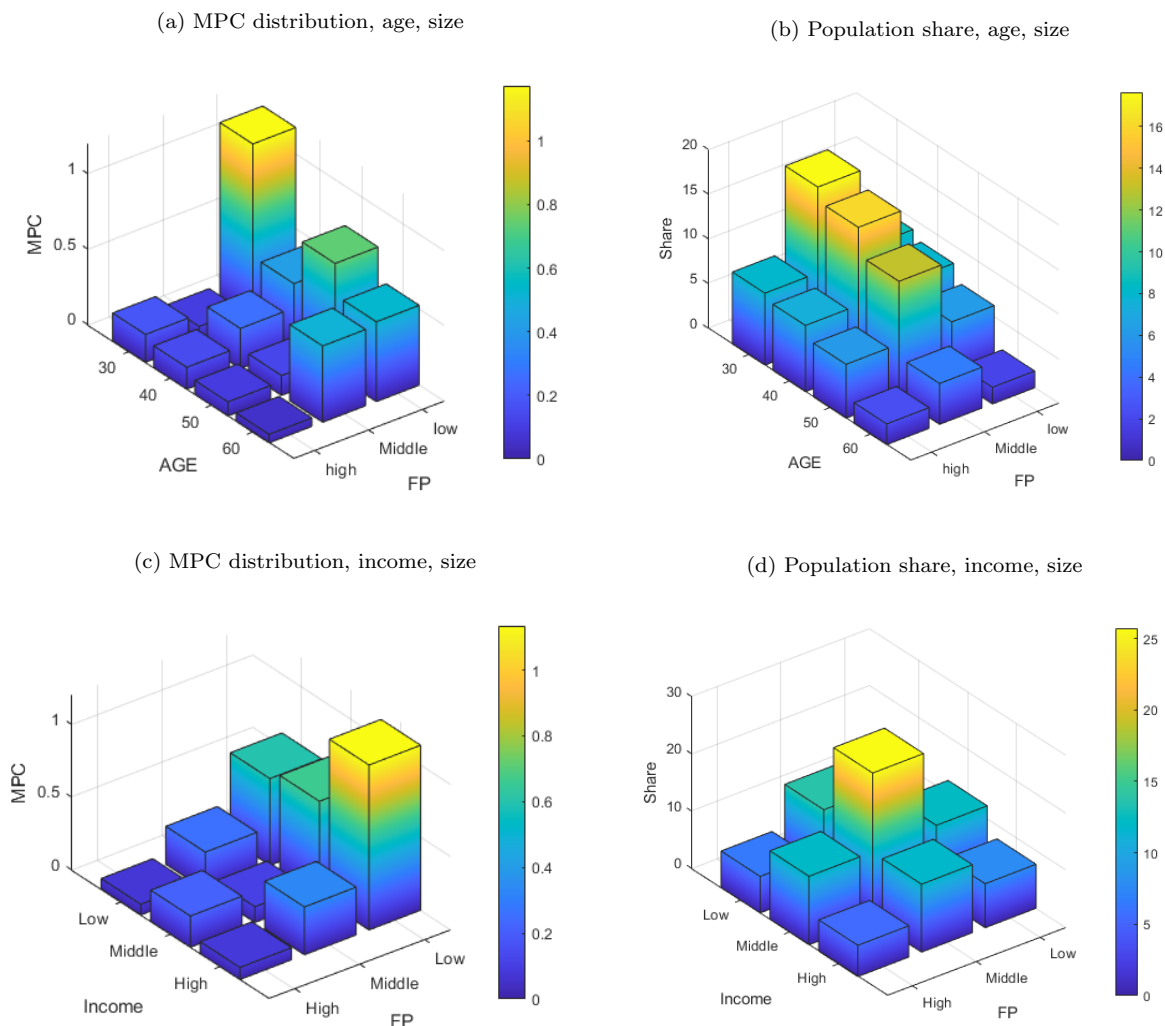


Notes: Panel (a) plots absolute final payment size against quarterly before-tax income, and Panel (b) plots the payment-to-income ratio against quarterly before-tax income. The solid line in each panel shows the fitted linear regression.

F Conditional MPC Heterogeneity

F.1 Consumption Response by the Absolute Payment Size

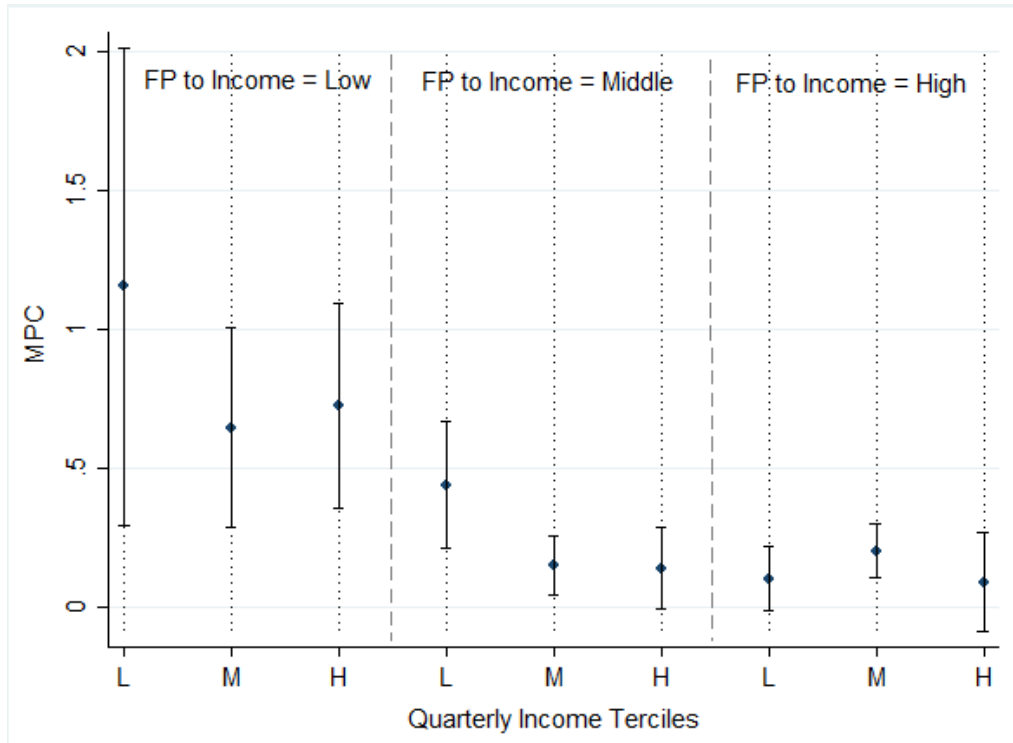
Figure F.1: MPC Heterogeneity by Payment Size (Level)



Notes: Left panels plot conditional MPC estimates and right panels plot population shares across absolute payment size terciles (low, middle, high), by age group (panels a and b) and income tercile (panels c and d). Bar color reflects the magnitude of the MPC or population share as indicated by the respective color scales. Standard errors are clustered at the individual level.

F.2 MPC by Relative Size conditional on Income

Figure F.2: Conditional MPC Heterogeneity by Relative Size

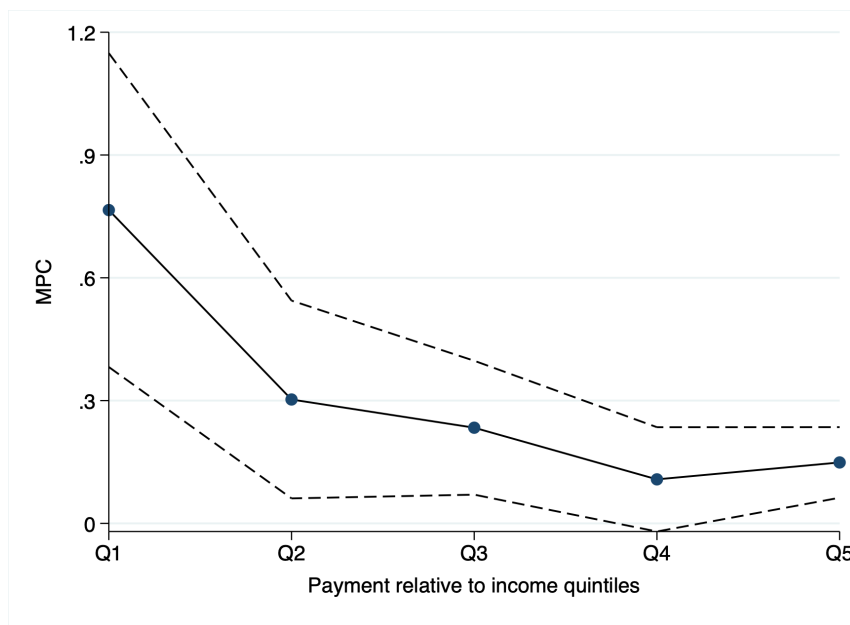


Notes: This figure plots MPC estimates across quarterly income terciles (L = low, M = middle, H = high) within each payment-to-income tercile. Dots denote point estimates and bars denote 95 percent confidence intervals with standard errors clustered at the individual level.

G Robustness Analysis

G.1 Alternative Grouping

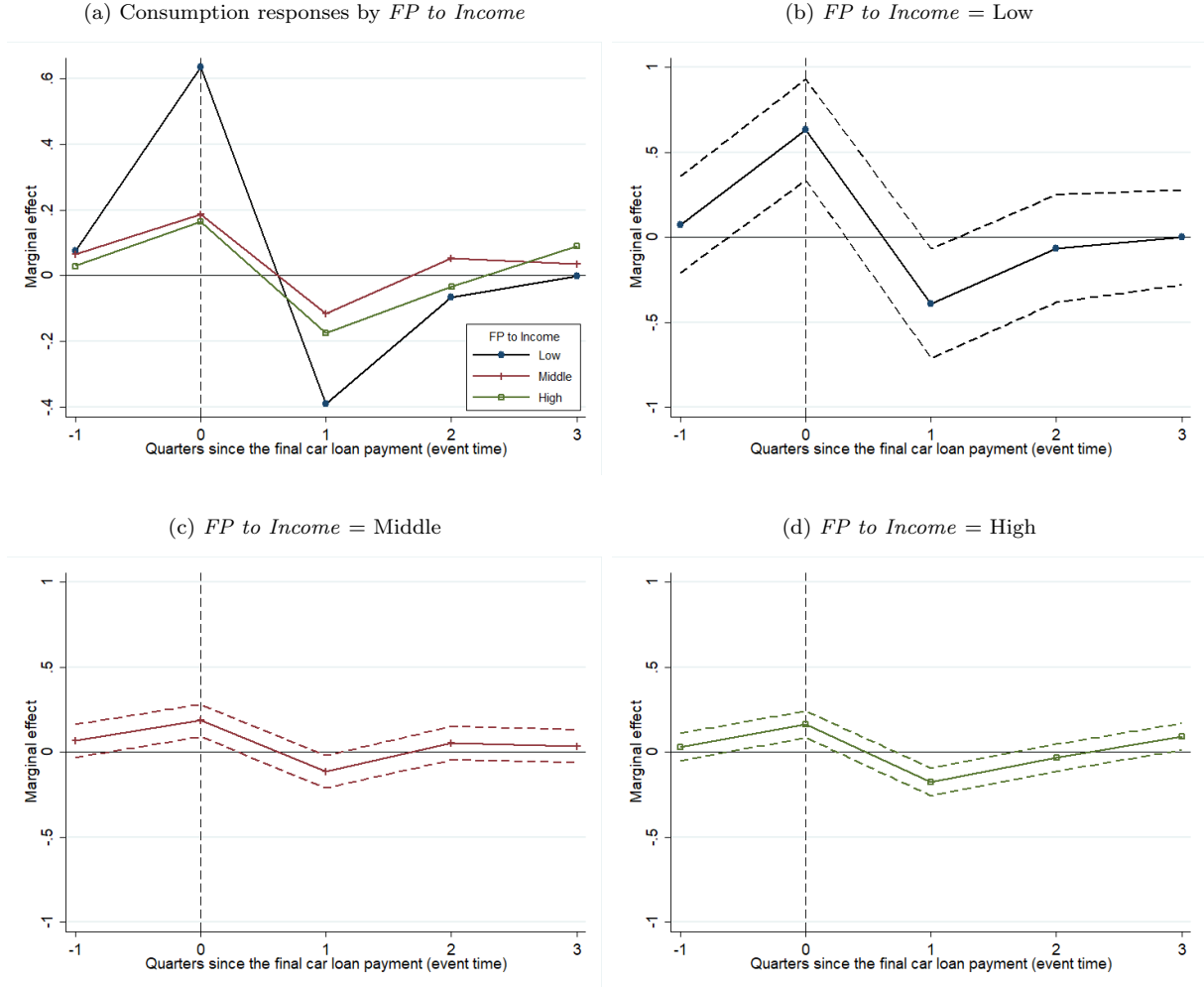
Figure G.1: Effects by Payment Size Relative to Income (FP to Income) Quintiles



Notes: This figure plots MPC estimates across five quintiles of the payment-to-income ratio. The solid line denotes point estimates and dashed lines denote 95 percent confidence intervals with standard errors clustered at the individual level.

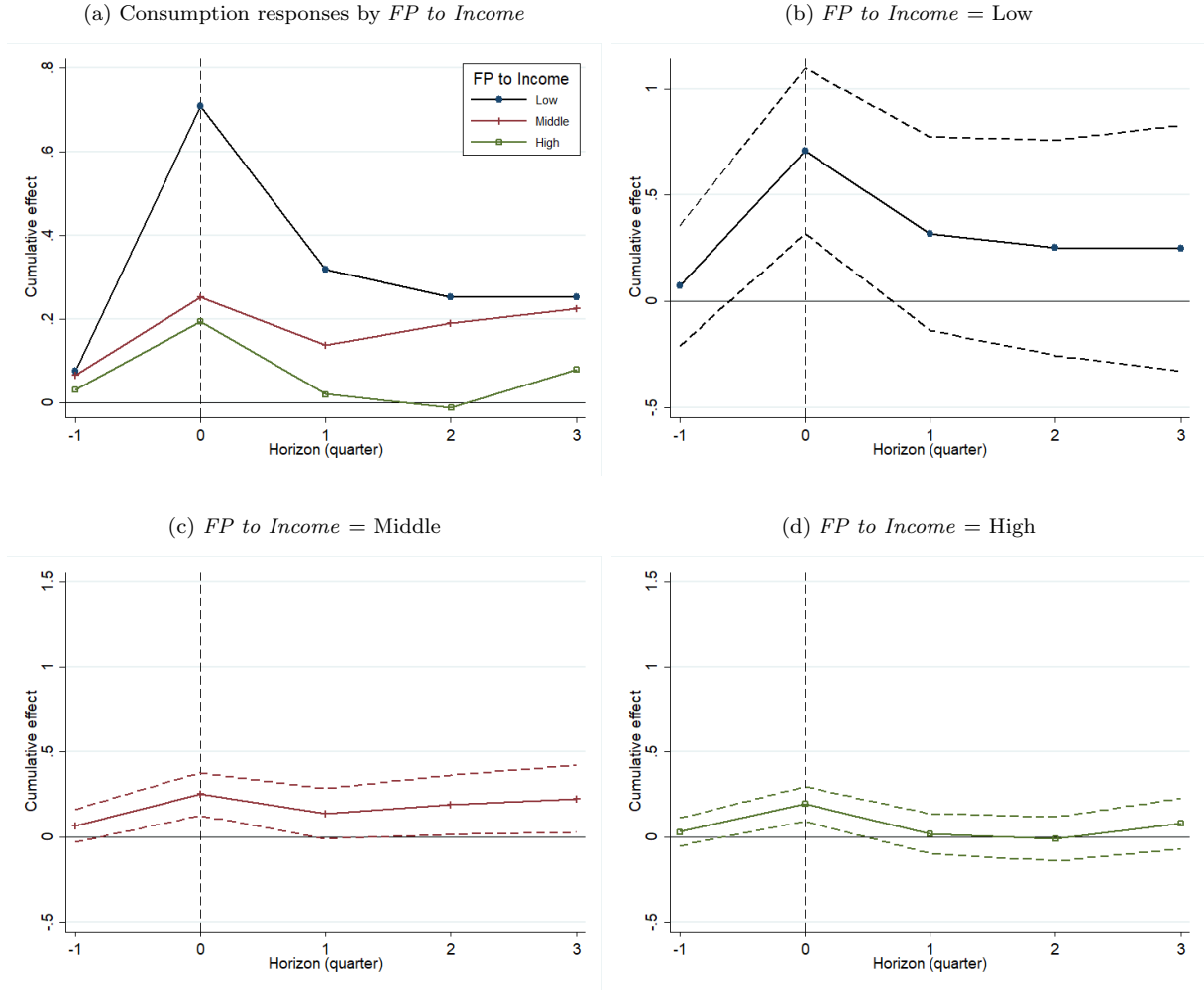
G.2 Heterogeneity in Consumption Dynamics

Figure G.2: Consumption Dynamics (**marginal**) by Payment Size Relative to Income



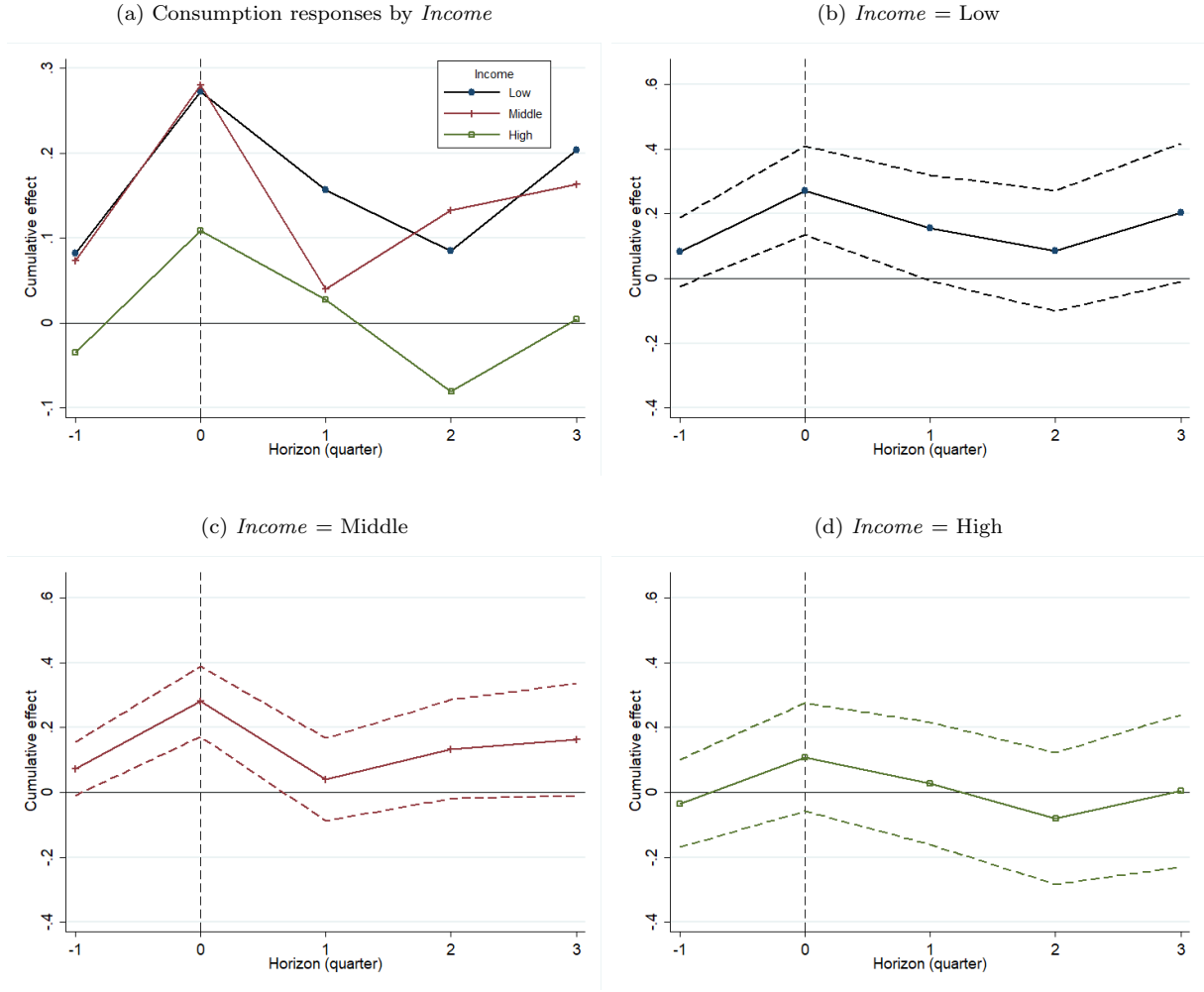
Notes: This figure plots the estimated marginal consumption response across payment-to-income terciles. Panel (a) overlays all three tercile groups, and panels (b) through (d) display results separately for the low, middle, and high terciles. The horizontal axis measures quarters relative to loan completion. Solid lines denote point estimates and dashed lines denote 95 percent confidence intervals with standard errors clustered at the individual level.

Figure G.3: Consumption Dynamics (**cumulative**) by Payment Size Relative to Income



Notes: This figure plots the estimated cumulative consumption response across payment-to-income terciles. Panel (a) overlays all three tercile groups, and panels (b) through (d) display results separately for the low, middle, and high terciles. The horizontal axis measures quarters relative to loan completion. Solid lines denote point estimates and dashed lines denote 95 percent confidence intervals with standard errors clustered at the individual level.

Figure G.4: Consumption Dynamics (**cumulative**) by Income



Notes: This figure plots the estimated cumulative consumption response across quarterly income terciles. Panel (a) overlays all three income groups, and panels (b) through (d) display results separately for the low, middle, and high income terciles. The horizontal axis measures quarters relative to loan completion. Solid lines denote point estimates and dashed lines denote 95 percent confidence intervals with standard errors clustered at the individual level.

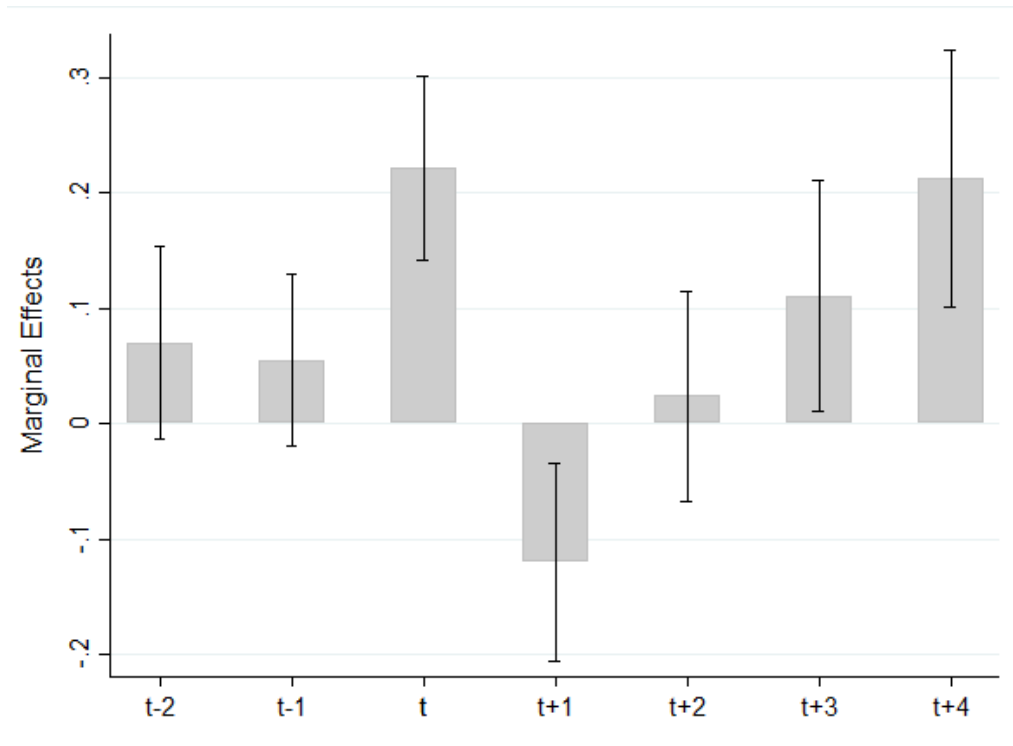
G.3 Estimation Results in Korean WON

Table G.1: Excess Sensitivity

	(1)	(2)	(3)	(4)	(5)	(6)
	$\Delta C_{i,t}$	$\Delta C_{i,t}$	$\Delta \ln C_{i,t}$	$\Delta C_{i,t}$	$\Delta C_{i,t}$	$\Delta \ln C_{i,t}$
FP	0.196*** (0.028)	0.179*** (0.028)		0.203*** (0.029)	0.177*** (0.028)	
FP to Income			0.350*** (0.044)			0.357*** (0.045)
Constant	0.232 (0.429)	0.809 (0.530)	0.022* (0.011)	0.104 (0.489)	2.461** (1.198)	0.049** (0.025)
Control Variables	×	○	○	×	○	○
Time and Region FE	○	○	○	○	○	○
Individual FE	×	×	×	○	○	○
R^2	0.000	0.023	0.024	0.002	0.02	0.021
Observations	141,933	141,933	141,933	141,933	141,933	141,933

Notes. FP and FP to Income indicates the final car loan payment and the payment relative to quarterly income. Control variables include the changes in income, annual income level, the changes in credit card limits, credit card utilization rates, credit grades, debt to income ratios and age dummies (30-39, 40-49, 50-59, 60-69 and 70+). Considering the measurement errors, observations with final payments to quarterly income greater than 1.5 were excluded from the sample. Robust standard errors in parentheses are clustered at the individual level. *, ** and *** represent significance at the 10%, 5% and 1%, respectively.

Figure G.5: Marginal Effects on Marginal Propensity to Consume



Notes: This figure plots leads and lags of the marginal consumption response estimated from equation (1) in Korean won, with an extended event window of two pre-period leads and four post-period lags. The period t denotes the quarter of loan completion. Bars denote point estimates and lines denote 95 percent confidence intervals with standard errors clustered at the individual level.

H Welfare Loss Analysis

To derive the potential welfare loss of deviating from the consumption smoothing behavior, we first define the optimal consumption decision under the life-cycle permanent income hypothesis. Consider the optimal consumption plan, c_{i,w_t}^{pih} , where each individual maximizes the life-time utility $U(c) = \sum_t \gamma^t u(c_t)$ given wealth w and prices p as follows:

$$c_{i,w_t}^{pih} = \arg \max_{c_t} \{U(c_t) \text{ s.t. } p_{t+1}c_t \leq w\} \quad (\text{H.1})$$

where $p_{t+1}c_t = \sum_t \frac{c_{i,t}^{pih}}{R^t}$ and $U(c) = \sum_t \beta^t u(c_{i,t})$. By the envelope theorem, we get

$$U(c_w^{pih}) - U(c_w^{deviate}) \approx -\frac{1}{2} \gamma^t \cdot \frac{\partial^2 u(c_t^{pih})}{\partial c^2} \cdot (c_t^{pih})^2 \cdot \left(\frac{c_t^{deviate} - c_t^{pih}}{c_t^{pih}} \right)^2 \quad (\text{H.2})$$

We use the amount of wealth, $\tilde{w}$, for each individual to keep at the utility level under $c_w^{deviate}$ to get the value function as follows:

$$U(c_w^{pih}) - U(c_{\tilde{w}}^{pih}) \approx -\left(\frac{\tilde{w} - w}{w} \right) \sum_t \gamma^t \cdot \left(\frac{\partial u(c_t^{pih})}{\partial c} \cdot c_t^{pih} \right) \quad (\text{H.3})$$

For simplicity, we consider $\gamma = 1$. Then, combining above two equations gives the potential welfare loss function (i.e., equation 5).

$$Welfare \text{ loss } (c_i^{deviate}, c_i^{pih}) \approx \frac{\delta}{2} \cdot \sum_t \zeta_t \left(\frac{c_t^{deviate} - c_t^{pih}}{c_t^{pih}} \right)^2 \quad (\text{H.4})$$

where δ captures the curvature of the utility function. ζ_t is the utility weight function where $\zeta_t = \gamma^t \frac{\partial u(c_t^{pih})}{\partial c} c_t^{pih} / \sum_i \gamma^n \frac{\partial u(c_n^{pih})}{\partial c} c_n^{pih} = \frac{\gamma^t u(c_t^{pih})}{U(c^{pih})}$ as we assume the utility function $u(c) = c^{1-\delta}/(1-\delta)$.

The consumption plan at time t , c_t , is defined as

$$c_t = \begin{cases} c^{pih} & \text{without predictable income changes} \\ c^{pih} + MPC \cdot FP & \text{with predictable income changes} \end{cases} \quad (\text{H.5})$$

where FP indicates the amount of predictable income changes following the final car loan payment at time t . Then, the deviation from the optimal consumption plan is defined as $c_t^{deviate}$ where

$$\frac{c_t^{deviate} - c_t^{pih}}{c_t^{pih}} = \begin{cases} 0 & \text{without predictable income changes} \\ \frac{(1-MPC) \cdot FP}{c^{pih}} & \text{with predictable income changes} \end{cases} \quad (\text{H.6})$$

and therefore, the welfare loss from deviation becomes

$$Welfare\ loss\ (c_i^{deviate}, c_i^{pih}) \approx \frac{\delta}{2} \cdot \left(\frac{(1 - MPC) \cdot FP}{c^{pih}} \right)^2 \quad (\text{H.7})$$

where c^{pih} is equal to permanent income and FP/c^{pih} represents the final payment size relative to one's quarterly income.